THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – I : Introduction & Basic Concepts

★ Note: Syllabus: System, boundary, surroundings, control volume, universe, types of systems, macroscopic & microscopic views, continuum, equilibrium, state, property, process, cycle, reversibility, quasi-static & irreversible processes, causes of irreversibility

1. What is Thermodynamics?

Thermodynamics is the science of energy – its storage, transformation (heat ⇄ work) and the properties of matter involved. The word comes from Greek: therme (heat) + dynamis (power).

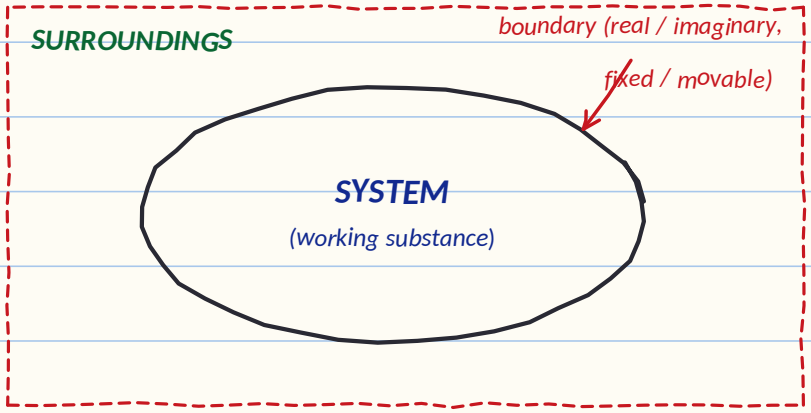
- Deals with energy interactions between a system and its surroundings, not with the rate (that is heat transfer).
- Based on experimental laws – Zeroth, First, Second and Third – which have never been violated.
- Applications: IC engines, steam & gas turbines, refrigerators, air-conditioners, compressors, rockets, power plants.
- Two approaches: macroscopic (classical) and microscopic (statistical).

2. System, Boundary, Surroundings, Universe

- System : a definite quantity of matter or a region in space chosen for study.
- Surroundings : everything outside the system that can interact with it.
- Boundary : real or imaginary surface separating system and surroundings. It may be fixed or movable. It has zero thickness and zero

volume.

- Universe : System + Surroundings.



Universe = System + Surroundings

Control volume : fixed region of space

Control surface : boundary of the CV

Boundary : thickness = 0, no volume

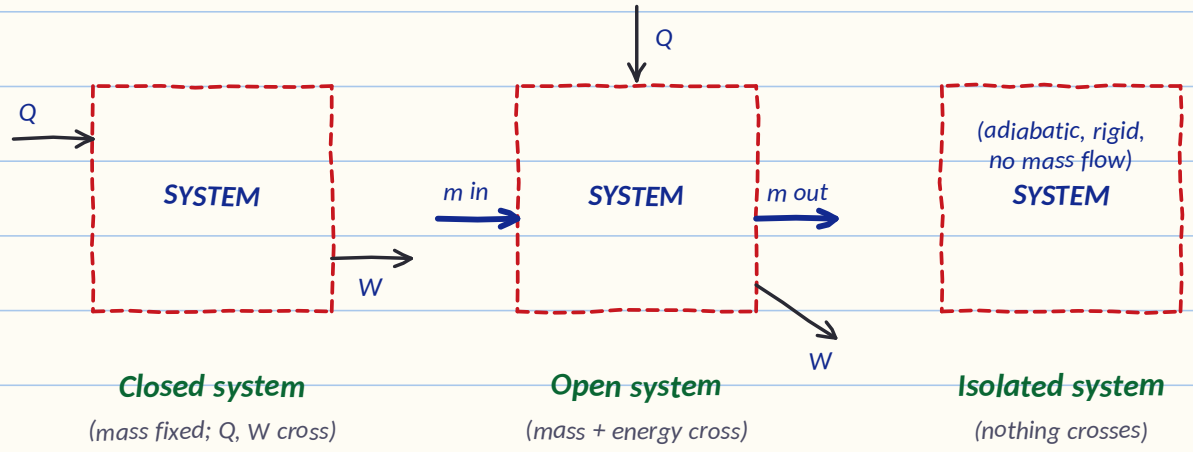
Types: closed, open, isolated

Control volume & control surface

A control volume is a fixed region in space (e.g. a turbine, nozzle, compressor) across which mass can flow. Its boundary is called the control surface. Used for open-system (flow) analysis.

3. Types of Systems

Type	Mass transfer	Energy transfer	Example
Closed (control mass)	No	Yes (Q, W)	Gas in piston-cylinder
Open (control volume)	Yes	Yes	Turbine, compressor, nozzle
Isolated	No	No	Thermos flask (ideal), universe



- Adiabatic system: no heat crosses the boundary ($Q = 0$). Rigid system: no work crosses ($W = 0$).
- Homogeneous system: single phase. Heterogeneous system: more than one phase (e.g. ice + water).

4. Macroscopic and Microscopic Viewpoints

Macroscopic (classical)	Microscopic (statistical)
Considers the matter as a whole	Considers molecules individually
No assumption about structure	Needs model of molecular structure
Few properties: P, V, T measured	Many variables: velocity, position of each
Simple, uses senses/instruments	Complex, uses statistics & probability
Properties are averages of molecules	Behaviour of each molecule is studied

★ Note: In this course we use the macroscopic approach. It works because a system consists of a very large number of molecules, so the average behaviour is steady and measurable.

5. Concept of Continuum

Matter is made of discrete molecules with empty space between. Classical thermodynamics ignores this and treats matter as a continuous medium (continuum) so that properties such as density, pressure and temperature vary smoothly from point to point.

- Density at a point: $\rho = \lim (\delta m/\delta V)$ as $\delta V \to \delta V'$, where $\delta V'$ is the smallest volume that still contains enough molecules to give a statistical average.

$$\rho = \lim \delta m / \delta V \quad (\delta V \to \delta V')$$

- Valid when the mean free path of molecules $\ll$ characteristic dimension of the system.
- Fails in highly rarefied gases (very high vacuum, upper atmosphere) – there the Knudsen number is large.

6. Thermodynamic Property, State, Path

- Property : any observable characteristic of a system (P, V, T, U, H, S ...). It depends only on the state, not on how the state was reached.
- Intensive properties: independent of mass (P, T, density, specific volume). Extensive properties: depend on mass (V, U, H, S, m).
- Specific property = extensive / mass, e.g. $v = V/m$, $u = U/m$. Specific extensive property behaves like an intensive one.
- State : the condition of a system described by its properties. Two independent properties fix the state of a simple compressible pure substance.
- Path : the series of states passed through. Process : change of state from one equilibrium state to another. Cycle : a series of processes after which

the system returns to its initial state.

★ Note: Heat and work are NOT properties – they are path functions. dQ, dW are inexact differentials; dU, dH, dS are exact (point functions).

7. Thermodynamic Equilibrium

A system is in thermodynamic equilibrium when its properties do not change with time and there is no spontaneous change when it is isolated. It requires three types of equilibrium to hold together:

→ Mechanical equilibrium – no unbalanced force; pressure uniform (no pressure gradient).

→ Thermal equilibrium – no temperature difference; temperature uniform.

→ Chemical equilibrium – no chemical reaction, no mass transfer by concentration difference.

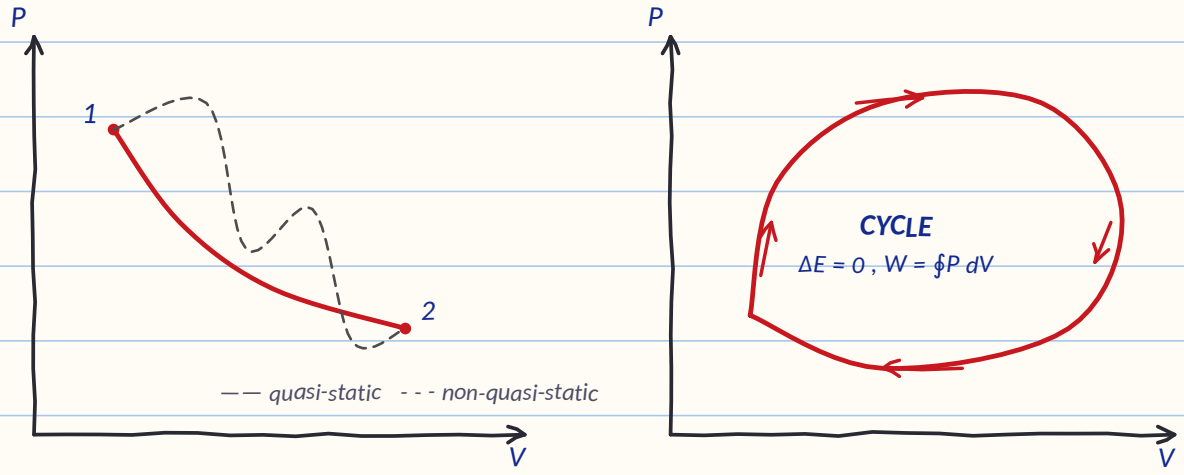
- Only equilibrium states can be represented as points on a property diagram (P-V, T-S etc.).
- Phase equilibrium is also included when more than one phase is present.

8. Processes and Cycles

Common processes on a P-V diagram (n = polytropic index, $PV^n = C$ written as $PV^n = C$):

Process	Constant	Index n	Work 1→2
Isobaric	P	0	P(V2 – V1)
Isochoric	V	∞	0
Isothermal	T (PV)	1	P1V1 ln(V2/V1)
Adiabatic	Q = 0	γ	(P1V1 – P2V2)/(γ–1)
Polytropic	PV^n	n	(P1V1 – P2V2)/(n–1)

Cycle: process 1-2-3-4-1 returns to initial state, so ΔE = 0 and net work = net heat = area enclosed. Clockwise on P-V = power cycle (engine); anticlockwise = refrigeration / heat-pump cycle.



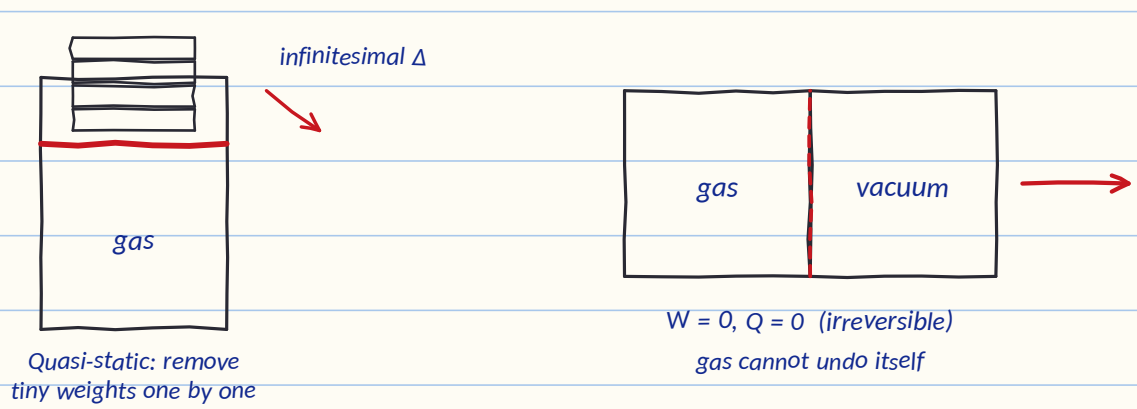
9. Quasi-static (Quasi-equilibrium) Process

A process that proceeds so slowly that the system passes through a continuous series of equilibrium states – each intermediate state is infinitesimally close to equilibrium.

- Example: gas in a cylinder with a piston carrying many small weights – remove one tiny weight at a time and let the gas settle.
- Can be drawn as a continuous line on a property diagram; a non-quasi-static process can only be shown with dotted line joining end

points.

- It is an idealisation. Real processes are fast, but many (like slow engine strokes) approximate it. It gives the maximum work output for expansion and minimum work input for compression.



10. Reversible and Irreversible Processes

Reversible process

A process which can be reversed so that the system and surroundings both return exactly to their initial states, leaving no trace in the universe.

- Must be quasi-static AND free from friction, free expansion, finite temperature-difference heat transfer, mixing etc.
- Every reversible process is quasi-static, but not every quasi-static process is reversible (e.g. quasi-static with friction).
- Gives maximum efficiency; used as the ideal standard (Carnot cycle).

Irreversible process

A process that cannot be reversed without leaving some change in the system or surroundings. All natural / real processes are irreversible.

- Examples: free expansion, heat flow from hot to cold body, mixing of two gases, combustion, flow through a valve (throttling), friction, electrical

resistance heating.

11. Causes of Irreversibility

Cause	Explanation / example
Friction	Mechanical (solid-solid), fluid friction – work turned to heat
Unrestrained expansion	Free expansion of gas into vacuum
Heat transfer through finite ΔT	Heat from a hot to a cold body
Mixing	Two different fluids / gases mixing
Chemical reactions	Combustion, diffusion, dissolving
Electrical effects	I^2R loss, hysteresis, magnetization losses
Inelastic deformation	Plastic deformation of solids
Non-quasi-static process	Finite pressure / temperature gradients

★ Note: Internal irreversibility occurs within the system (friction, turbulence, free expansion). External irreversibility occurs at the boundary (heat transfer through finite temperature difference).

12. Units, Pressure and Temperature

- SI base units: length m, mass kg, time s, temperature K, amount of substance mol.
- Force: 1 N = 1 kg·m/s². Pressure: 1 Pa = 1 N/m². Energy: 1 J = 1 N·m. Power: 1 W = 1 J/s.

$1\ bar = 10^5\ Pa = 100\ kPa \quad 1\ atm = 1.01325\ bar = 760\ mm\ Hg$

$1\ kWh = 3600\ kJ \quad 1\ kJ/kg = 1000\ m^2/s^2$

Types of pressure

- Absolute pressure = pressure measured from perfect vacuum (zero). Used in all thermodynamic equations.

- Gauge pressure = pressure above atmosphere; vacuum pressure = pressure below atmosphere.

$$P_{abs} = P_{atm} + P_{gauge}$$

$$P_{abs} = P_{atm} - P_{vacuum}$$

- Manometer / barometer: pressure of a fluid column $P = \rho g h$ (ρ of mercury = 13600 kg/m³).

Temperature scales

$$K = °C + 273.15 \qquad °F = 1.8\,°C + 32$$

$$°R = °F + 459.67 = 1.8\,K$$

- Zeroth law gives the basis of temperature measurement. Fixed point: triple point of water = 273.16 K (0.01 °C).
- Thermometers: liquid-in-glass (mercury, alcohol), constant-volume gas, resistance (RTD), thermocouple, optical pyrometer.

13. Equation of State and State Postulate

The relation among P, V and T of a substance is its equation of state. For an ideal (perfect) gas:

$$P V = m R T \quad \text{or} \quad P v = R T \quad \text{or} \quad P V = n R_u T$$

- $R = R_u/M$, universal gas constant R_u = 8.314 kJ/kmol·K. For air R = 0.287 kJ/kg·K.
- Gas laws: Boyle (T const: PV = C), Charles (P const: V/T = C), Gay-Lussac (V const: P/T = C), Avogadro (equal volumes, same P & T, same number of molecules).
- Real gases deviate at high pressure / low temperature: van der Waals equation $(P + a/v^2)(v - b) = RT$.

★ Note: State postulate: the state of a simple compressible system is completely fixed by two independent intensive properties. Example: (P, T) or (P, v) or (T, v).

Point function vs Path function

Point function	Path function
Depends only on end states	Depends on the path followed
Exact differential (dU, dP, dV)	Inexact differential (δQ, δW)
Cyclic integral $\oint dX = 0$	Cyclic integral $\oint \delta Q$, $\oint \delta W \neq 0$
Examples: P, T, V, U, H, S	Examples: heat, work

14. Solved Numerical Problems

Problem 1 : Absolute pressure

A pressure gauge on a tank reads 2.5 bar. Barometer reads 760 mm Hg. Find absolute pressure.

$$P_{abs} = 2.5 + 1.01325 = 3.513 \text{ bar}$$

Problem 2 : Vacuum gauge

A condenser vacuum gauge reads 400 mm Hg when barometer reads 760 mm Hg. Find the absolute pressure.

$$P_{abs} = 760 - 400 = 360 \text{ mm Hg} = (360/760)(1.01325) = 0.48 \text{ bar}$$

Problem 3 : Manometer

A U-tube mercury manometer shows a difference of 0.2 m. Find the gauge pressure.

$$\Delta P = \rho g h = 13600 \times 9.81 \times 0.2 = 26\,683 \text{ Pa} \approx 26.7 \text{ kPa}$$

Problem 4 : Isothermal expansion work

1 kg of air expands isothermally at 300 K from 5 bar to 1 bar. Find the work done ($R = 0.287$ kJ/kg·K).

$$W = m\,R\,T\,\ln(P_1/P_2) = 1 \times 0.287 \times 300 \times \ln 5$$

$$W = 86.1 \times 1.609 = 138.6\ kJ$$

Problem 5 : Polytropic expansion

A gas at 4 bar, 0.02 m^3 expands as $PV^{1.3} = C$ to 0.06 m^3. Find final pressure and work.

$$P_2 = P_1(V_1/V_2)^{1.3} = 4\ (1/3)^{1.3} = 0.959\ bar$$

$$W = (P_1V_1 - P_2V_2)/(n-1) = (8000 - 5756)/0.3 = 7481\ J \approx 7.48\ kJ$$

Problem 6 : Net work of a rectangular cycle

Cycle: 1-2 isobaric at 2 bar, V from 0.1 to 0.3 m^3; 2-3 isochoric to 1 bar; 3-4 isobaric back to 0.1 m^3; 4-1 isochoric.

$$W_{12} = 2\times10^5(0.2) = +40\ kJ\ ;\quad W_{34} = 1\times10^5(-0.2) = -20\ kJ$$

$$W_{net} = 40 - 20 = 20\ kJ\ \ (= \text{enclosed area})$$

15. Quick Revision – Important Points

- System + Surroundings = Universe; boundary has zero thickness.
- Closed: mass fixed. Open: mass crosses. Isolated: neither mass nor energy.
- Property = point function (exact differential); heat & work = path functions.
- Equilibrium = mechanical + thermal + chemical (+ phase).
- Quasi-static ⇒ infinite slowness; reversible ⇒ quasi-static + no dissipation.
- Cycle: $\oint dP = \oint dV = \oint dT = 0$ for any property; $\oint dQ = \oint dW$.

Frequently asked questions

→ Differentiate between closed and open systems with examples.

→ Explain quasi-static process with a neat sketch. How is it different from

reversible?

→ What are the causes of irreversibility? Explain with examples.

→ Explain thermodynamic equilibrium and its types.

→ State the concept of continuum. When does it fail?

16. Short Answer Questions (Exam Ready)

→ Define thermodynamics. Science of energy and its transformations between heat and work, and the properties of the matter involved.

→ What is a thermodynamic property? A point function describing the state, independent of the path (P, T, V, U, H, S).

→ Intensive vs extensive? Intensive – independent of mass (P, T); extensive – proportional to mass (V, U, H).

→ What is a cycle? A series of processes that returns the system to its initial state; $\oint dX = 0$ for properties.

→ Why is free expansion irreversible? The gas cannot return to the original container without external work, and no work is obtained during expansion.

→ State the zeroth requirement for equilibrium. Mechanical, thermal and chemical equilibrium must exist together.

→ Is every quasi-static process reversible? No. A quasi-static process with friction is not reversible.

→ Examples of open systems? Boiler, turbine, nozzle, compressor, pump, heat exchanger.

~ End of Unit I ~

THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – II : First Law of Thermodynamics

★ Note: Syllabus: Energy in state and in transition, work and heat, point and path function, zeroth law, PMM-I, Joule's experiment, first law and applications, limitations of first law, enthalpy, thermal reservoir, heat engine, heat pump, parameters of performance

1. Energy in State and Energy in Transition

Energy is the capacity to do work. It exists in two ways:

- Stored energy (energy in state) – a property of the system: kinetic energy $KE = \frac{1}{2}mV^2$, potential energy $PE = mgz$, and internal energy U.
- Energy in transition – crosses the system boundary only while a process occurs: heat (Q) and work (W). They are not stored in the system.

$$\text{Total energy } E = U + KE + PE = U + \tfrac{1}{2} m V^2 + m g z$$

Internal energy (U)

- Energy of molecules: translational, rotational, vibrational kinetic energy + intermolecular potential energy + chemical & nuclear energy.
- Extensive property, unit kJ. Specific internal energy $u = U/m$ (kJ/kg).
- For an ideal gas U depends only on temperature: $u = f(T)$, $du = c_v\, dT$.

2. Work

Work is energy transfer across the boundary due to a force acting through a distance – or, thermodynamically, any interaction other than temperature difference that can be fully converted to lifting a weight.

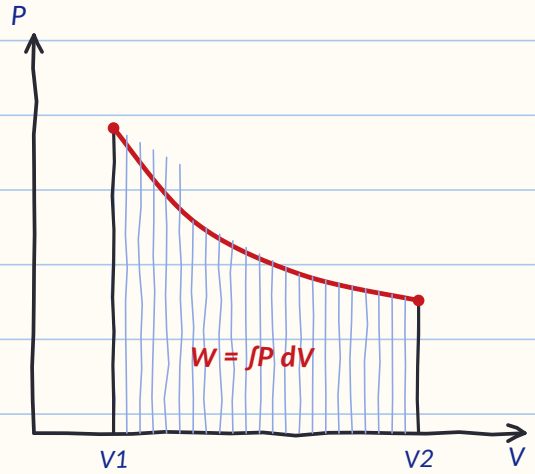
- Sign convention: work done BY the system = positive (+W); work done

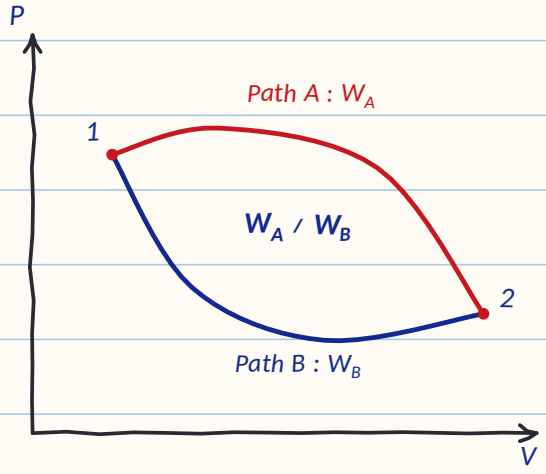
ON the system = negative (–W).

- Unit: J, kJ. Power = work per unit time (W).

Displacement (PdV) work

$$\delta W = P\,dV \qquad W_{1\text{-}2} = \int P\,dV \ \text{(from } V_1 \text{ to } V_2\text{)}$$





- Work = area under the P-V curve, valid only for quasi-static processes.

  Area depends on the path ⇒ work is a path function.

- Expansion: dV > 0, W positive. Compression: dV < 0, W negative.

Other modes of work

Mode	Expression
Shaft work	$W = T\,\omega$ (torque × angular velocity)
Electrical work	$W = V\,I\,t$
Spring work	$W = \frac{1}{2}\,k\,(x_2^2 - x_1^2)$
Stretching of wire	$W = \int \sigma\,A\,dx$
Surface film	$W = -\int \sigma\,dA$

3. Heat

Heat is energy in transit across the boundary because of a temperature difference. It flows from high to low temperature.

- Modes of heat transfer: conduction, convection, radiation.
- Sign: heat added TO the system = positive (+Q); heat rejected BY the system = negative (−Q).
- Adiabatic process: Q = 0. Heat transfer rate $\dot{Q}$ in W; specific heat transfer q = Q/m.

Heat vs Work – Comparison

Heat	Work
Due to temperature difference	Due to force × displacement
Disorganised molecular energy	Organised energy
Low-grade energy	High-grade energy
Cannot be fully converted to work	Fully convertible to heat
Both are path functions, boundary phenomena	Both are transient, not stored

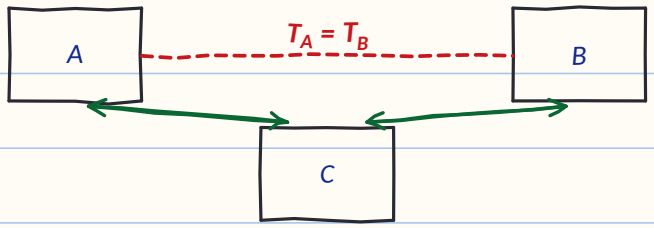
4. Point Function and Path Function

- Point function: value depends only on the end states – properties (P, T, V, U, H, S). Exact differential; $\oint dX = 0$.
- Path function: value depends on the path between end states – heat and work. Inexact differential δ; $\oint \delta Q \neq 0$, $\oint \delta W \neq 0$.

$$\int dU = U_2 - U_1 \quad \text{but} \quad \int \delta W = W_{1\text{-}2} \ (\text{NOT } W_2 - W_1)$$

5. Zeroth Law of Thermodynamics

★ Note: If two bodies are each in thermal equilibrium with a third body, then they are in thermal equilibrium with each other. (R.H. Fowler, 1931)



$T_A = T_C$ and $T_B = T_C$

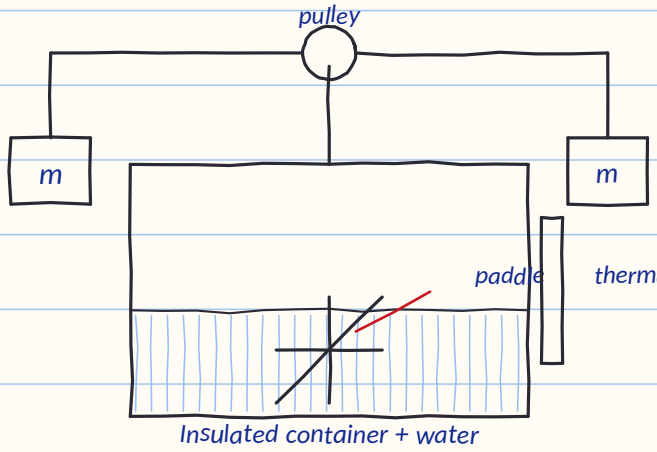
$\Rightarrow T_A = T_B$

(basis of thermometry)

- Basis for the concept of temperature – bodies in thermal equilibrium have the same temperature.
- Makes thermometers possible: thermometer (body C) reading equal for A and B $\Rightarrow T_A = T_B$.
- Called "zeroth" because it was recognised after the first and second laws but is logically more fundamental.

6. Joule's Experiment

Joule (1843-1850) converted known amounts of work into heat in an insulated vessel and measured the temperature rise of the water.



$W = 2\,m\,g\,h$

→ raises T of water

1 cal = 4.186 J

$W \propto Q$ ($J = 4.186$)

→ Falling weights turn a paddle wheel which stirs the water; work done = $m\,g\,h$ (shaft work).

→ Water temperature rises; Q-equivalent = $m_w\,c_w\,\Delta T$. The container is insulated so no heat leaves.

→ Repeated with different liquids, paddle materials, masses – the ratio W/Q remained constant.

$W \propto Q \Rightarrow W = J\,Q$, $J = 4.186$ J/cal (mechanical equivalent of heat)

- Conclusion: heat and work are mutually convertible forms of energy; in cyclic process $\oint \delta Q = \oint \delta W$ (in consistent units).

7. First Law of Thermodynamics

★ Note: Statement 1 (cycle): For a closed system undergoing a cyclic process, the net heat transfer equals the net work transfer: $\oint \delta Q = \oint \delta W$.

★ Note: Statement 2 (Clausius/Joule): Energy can neither be created nor destroyed, it can only be transformed from one form to another.

First law for a process (non-cyclic)

If a system goes from state 1 to 2 by path A and returns by path B: $\oint \delta Q = \oint \delta W \Rightarrow (Q_A - W_A) + (Q_B - W_B) = 0$. Different return paths give the same $Q-W$ $\Rightarrow Q-W$ is a property $= \Delta E$.

$$Q_{1\text{-}2} = E_2 - E_1 + W_{1\text{-}2}$$

$$\delta Q = dE + \delta W \qquad \text{(closed system)}$$

$$Q = \Delta U + W \qquad \text{(KE and PE changes neglected)}$$

- Isolated system: $Q = 0$, $W = 0 \Rightarrow \Delta E = 0$, energy of the universe is constant.
- $Q - W = \Delta E$ is also valid for irreversible processes; only the $P\,dV$ form needs quasi-static processes.

Perpetual Motion Machine of the First Kind (PMM-I)

A hypothetical machine that produces work continuously without any energy input (or delivers more energy than it receives). It violates the first law, hence impossible.

- Corollary: there is no PMM-I, $W_{cycle} = Q_{cycle}$ always.

Specific heats and ideal gas relations

$$c_v = (\partial u/\partial T)_v \qquad c_p = (\partial h/\partial T)_p$$

$$c_p - c_v = R \qquad \gamma = c_p/c_v$$

$$c_v = R/(\gamma-1) \qquad c_p = \gamma R/(\gamma-1)$$

- Air: $c_p = 1.005$, $c_v = 0.718$ kJ/kg·K, $R = 0.287$ kJ/kg·K, $\gamma = 1.4$.

8. Enthalpy

$$H = U + P\,V \qquad h = u + P\,v \quad (kJ/kg)$$

- Enthalpy is a property (U, P, V all are properties). Extensive; unit kJ.
- Physical meaning: total energy of a flowing fluid = internal energy + flow work (Pv).
- For constant-pressure quasi-static process in closed system: $Q = \Delta H$.
- Ideal gas: $h = f(T)$ only, $dh = c_p\,dT$.

9. First Law Applied to Closed Systems (Non-flow)

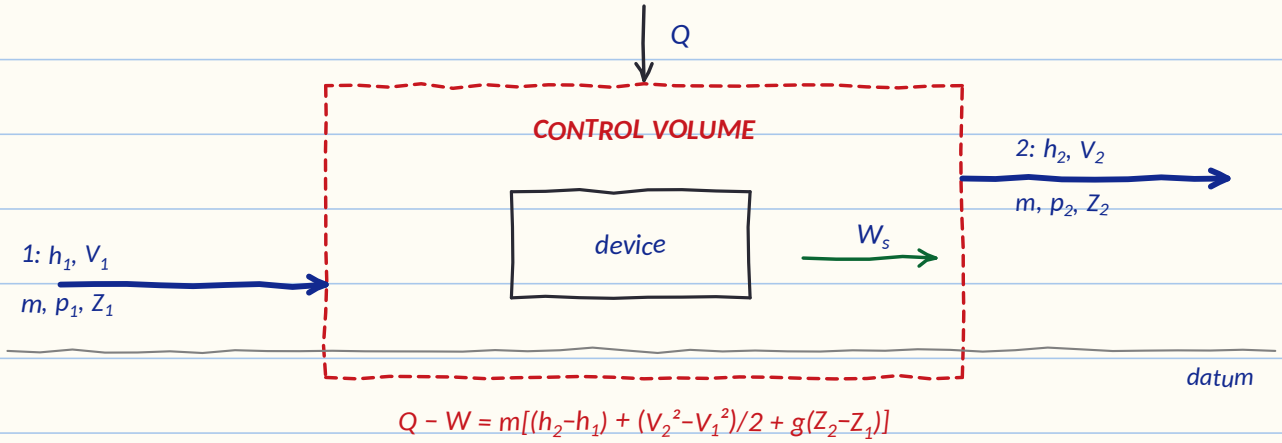
Process	W	ΔU	Q
Isochoric (V const)	0	$m\,c_v\Delta T$	ΔU
Isobaric (P const)	$P(V2-V1)=mR\Delta T$	$m\,c_v\Delta T$	$m\,c_p\Delta T = \Delta H$
Isothermal	$mRT\,\ln(V2/V1)$	0	W
Adiabatic (Q=0)	$(P1V1-P2V2)/(\gamma-1)$	−W	0
Polytropic PV^n=C	$(P1V1-P2V2)/(n-1)$	$m\,c_v\Delta T$	$W(\gamma-n)/(\gamma-1)$

$$T_2/T_1 = (V_1/V_2)^{\gamma-1} = (P_2/P_1)^{(\gamma-1)/\gamma} \quad \text{(adiabatic)}$$

- Free expansion: $Q = 0$, $W = 0 \Rightarrow \Delta U = 0$ (for ideal gas T unchanged).

10. First Law for Open Systems – SFEE

Steady Flow Energy Equation (SFEE) – steady state: mass flow in = mass flow out, properties at any point do not change with time, no accumulation of energy in the CV.



$$Q - W = m[(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(Z_2 - Z_1)]$$

$$\dot{m}_1 = \dot{m}_2 \rightarrow \rho_1 A_1 V_1 = \rho_2 A_2 V_2 \text{ (continuity)}$$

$$\dot{Q} - \dot{W}_s = \dot{m}\,[\,(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g\,(z_2 - z_1)\,]$$

- Per kg: $q - w = \Delta h + \Delta KE + \Delta PE$. Flow work = P v (included in enthalpy).

Applications of SFEE

Device	Assumptions	Result
Nozzle	q=0, w=0, Δz=0	$V2 = \sqrt{[2(h1-h2)+V1^2]}$
Diffuser	q=0, w=0	V2 reduces, h2 > h1
Turbine	q≈0, ΔKE,ΔPE≈0	w = h1 – h2
Compressor/Pump	q≈0	w = h1 – h2 (negative)
Throttle valve	q=0,w=0, ΔKE≈0	h1 = h2 (isenthalpic)
Boiler	w=0	q = h2 – h1
Condenser	w=0	q = h1 – h2 rejected
Heat exchanger	overall q=0	$\dot{m}_h \Delta h_h = \dot{m}_c \Delta h_c$

11. Limitations of the First Law

→ Does not tell the direction of a process – a cup of hot coffee cooling satisfies the law, but the reverse (coffee heating itself) does not occur.

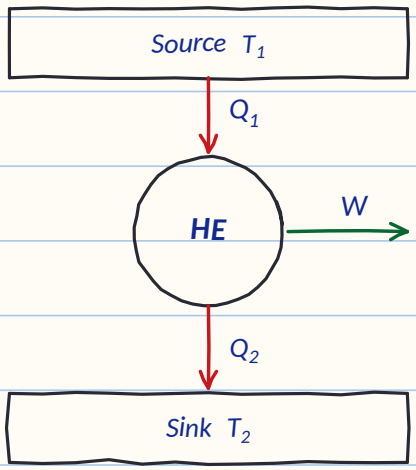
→ Does not tell whether a process is possible: all processes satisfying energy balance are allowed, yet many do not occur.

→ Does not distinguish between quality of energy: work and heat treated equal although heat cannot be completely converted to work.

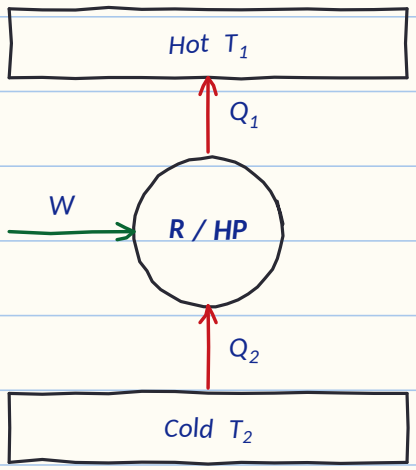
→ Does not give the limit of conversion of heat to work (efficiency) – needs the Second Law.

12. Thermal Reservoir, Heat Engine, Heat Pump

- Thermal energy reservoir (TER): very large body whose temperature does not change when heat is added or removed. Source supplies heat; sink absorbs heat.
- Heat engine: cyclic device that receives heat Q_1 from a high-T source, converts part to work W, and rejects Q_2 to a low-T sink.
- Refrigerator: device operating in cycle that extracts Q_2 from a cold space and rejects Q_1 to a warmer surrounding using work input. Heat pump: same cycle, but purpose is to supply Q_1 to the hot space.



Heat engine ($T_1 > T_2$)



Refrigerator / Heat pump

Parameters of performance

$$\eta_{th} = W_{net}/Q_1 = 1 - Q_2/Q_1$$

$$COP_R = Q_2/W = Q_2/(Q_1 - Q_2)$$

$$COP_{HP} = Q_1/W = Q_1/(Q_1 - Q_2)$$

$$COP_{HP} = COP_R + 1$$

- $\eta < 1$ always ($Q_2 \neq 0$); COP_R can be > 1; $COP_{HP} > 1$ always.
- 1 ton of refrigeration = 3.517 kW = 211 kJ/min (heat to freeze 1 ton of water at 0 °C in 24 h).
- Other measures: EER = capacity (Btu/h) / power (W); specific fuel consumption for engines.

13. Solved Numerical Problems

Problem 1

A system receives 100 kJ of heat and does 40 kJ of work. Find ΔU.

$$\Delta U = Q - W = 100 - 40 = 60 \text{ kJ}$$

Problem 2 : cycle of three processes

1-2: $Q = 40$, $W = 15$ kJ; 2-3: $Q = -35$, $\Delta U = -10$ kJ; 3-1: $W = -10$ kJ. Find the unknowns.

$$\Delta U_{12} = 40 - 15 = 25\ kJ\ ;\ W_{23} = Q - \Delta U = -35 + 10 = -25\ kJ$$

$$\Delta U_{31} = -(25 - 10) = -15\ kJ\ ;\ Q_{31} = -15 + (-10) = -25\ kJ$$

$$\Sigma Q = 40 - 35 - 25 = -20\ kJ = \Sigma W = 15 - 25 - 10 = -20\ kJ\ \checkmark$$

Problem 3 : constant pressure heating of air

1 kg air heated from 300 K to 400 K at constant pressure. Find Q, W, ΔU.

$$Q = c_p\Delta T = 1.005\times100 = 100.5\ kJ\ ;\ W = R\Delta T = 28.7\ kJ$$

$$\Delta U = c_v\Delta T = 0.718\times100 = 71.8\ kJ\ \ (= Q - W)$$

Problem 4 : adiabatic compression

1 kg air at 1 bar, 300 K is compressed adiabatically to 6 bar. Find T_2 and work.

$$T_2 = 300 \times 6^{0.2857} = 500.6\ K$$

$$W = c_v(T_1 - T_2) = 0.718 \times (-200.6) = -144.0\ kJ$$

Problem 5 : nozzle

Steam enters a nozzle with negligible velocity at h_1 = 3000 kJ/kg and leaves at h_2 = 2800 kJ/kg. Find exit velocity.

$$V_2 = \sqrt{[2(h_1 - h_2)]} = \sqrt{(2 \times 200 \times 1000)} = 632.5\ m/s$$

Problem 6 : steam turbine

$\dot{m}$ = 5 kg/s, h_1 = 3200, h_2 = 2400 kJ/kg, heat loss 40 kW. Find power output.

$$\dot{W} = \dot{m}(h_1 - h_2) - \dot{Q}_{loss} = 5 \times 800 - 40 = 3960\ kW$$

Problem 7 : air compressor

$\dot{m}$ = 0.5 kg/s, h_1 = 300, h_2 = 450 kJ/kg, heat loss 10 kW. Find power input.

$$\dot{W} = \dot{Q} - \dot{m}(h_2 - h_1) = -10 - 0.5 \times 150 = -85\ kW\ \ (85\ kW\ input)$$

Problem 8 : engine, refrigerator, heat pump

Engine: $Q_1 = 1000$, $W = 400$ kJ $\Rightarrow \eta = 40\,\%$, $Q_2 = 600$ kJ

Refrigerator: 10 kW cooling, COP = 4 $\Rightarrow$ $W = 2.5$ kW, $Q_1 = 12.5$ kW

Heat pump: $COP_{HP} = COP_R + 1 = 5$

14. Derivations to Remember

(a) SFEE derivation – steps

→ Take a control volume with inlet 1 and outlet 2, steady flow. Energy entering = energy leaving.

→ Energy carried by 1 kg of flowing fluid = $u + V^2/2 + gz$ plus flow work $P\,v$ needed to push it in.

$$\dot{m}(u_1 + P_1v_1 + V_1^2/2 + gz_1) + \dot{Q} = \dot{m}(u_2 + P_2v_2 + V_2^2/2 + gz_2) + \dot{W}_s$$

→ Since $h = u + Pv$, we get $\dot{Q} - \dot{W}_s = \dot{m}[(h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(z_2 - z_1)]$.

(b) Work in an adiabatic process

For $Q = 0$ the first law gives $W = -\Delta U = m\,c_v(T_1 - T_2)$. With $PV = mRT$ and $c_v = R/(\gamma-1)$:

$$W = m\,R\,(T_1 - T_2)/(\gamma - 1) = (P_1V_1 - P_2V_2)/(\gamma - 1)$$

(c) Work in an isothermal process

$$W = \int P\,dV = \int (mRT/V)\,dV = m\,R\,T\,\ln(V_2/V_1) = P_1V_1\,\ln(P_1/P_2)$$

(d) Relation $c_p - c_v = R$

For an ideal gas $h = u + RT \Rightarrow dh/dT = du/dT + R \Rightarrow c_p = c_v + R$.

15. Unsteady (Transient) Flow – Brief Idea

In filling or emptying a tank, the mass inside the control volume changes with time, so SFEE cannot be used. The energy balance over the process is:

$$Q - W = m_2u_2 - m_1u_1 + \Sigma\, m_eh_e - \Sigma\, m_ih_i$$

- Rigid insulated tank filled from a line at enthalpy h_i (no work, no heat): $m_2u_2 - m_1u_1 = m_ih_i$.
- For an ideal gas filling an evacuated tank: $T_{final} = \gamma\, T_{line}$. The flow work gets converted into internal energy.

16. More Solved Problems

Problem 9 : polytropic expansion of air

1 kg air expands polytropically ($n = 1.3$) from 5 bar, 400 K to 1 bar. Find W, ΔU and Q.

$$T_2 = 400\,(1/5)^{0.3/1.3} = 276\ K$$

$$W = R(T_1 - T_2)/(n-1) = 0.287 \times 124/0.3 = 118.6\ kJ$$

$$\Delta U = c_v(T_2 - T_1) = 0.718 \times (-124) = -89.1\ kJ$$

$$Q = \Delta U + W = +29.6\ kJ \quad \text{(heat is added)}$$

Problem 10 : stirring work in a rigid tank

A rigid insulated-then-leaky tank is stirred with 100 kJ of paddle work while 30 kJ of heat leaves. Find ΔU.

$$\Delta U = Q - W = (-30) - (-100) = +70\ kJ$$

Problem 11 : heat pump for heating

A house loses 30 kW; heat pump $COP_{HP} = 4$. Power required?

$$W = Q_1/COP_{HP} = 30/4 = 7.5\ kW \quad (22.5\ kW \text{ taken free from outside air})$$

Problem 12 : liquid heated in a heat exchanger

Water 2 kg/s, 20 → 60 °C ($c = 4.18$ kJ/kg·K):

$$\dot{Q} = \dot{m}\, c\, \Delta T = 2 \times 4.18 \times 40 = 334.4\ kW$$

17. Glossary & Short Answer Questions

→ Adiabatic wall – no heat crosses; diathermic wall – allows heat transfer.

→ Why is Q – W = ΔE valid for irreversible processes? It only balances energy, no assumption of reversibility.

→ Why is enthalpy useful? Combines U and flow work; Q = ΔH at constant P, and appears directly in SFEE.

→ Is the work in free expansion zero? Yes, because there is no resisting force (P_{ext} = 0).

→ Define heat pump COP. COP_{HP} = desired heating / work input = Q_1/W.

→ Is heat a property? No – a path function and energy in transit.

→ What does zeroth law establish? Temperature as a property and equality of temperature for thermal equilibrium.

18. Quick Revision & Important Questions

- Q – W = ΔE (closed); SFEE: q – w = Δh + ΔKE + ΔPE (open).
- Heat & work: path functions. U, H: properties. PMM-I impossible.
- h = u + Pv; c_p – c_v = R. Throttling is isenthalpic; nozzle increases KE.

→ State and explain the first law for a closed system undergoing a cycle and a process.

→ Describe Joule's experiment. What does it prove?

→ Derive the steady flow energy equation and apply to nozzle, turbine and compressor.

→ Define PMM-I. Why is it impossible?

→ What are the limitations of the first law? Differentiate heat engine, refrigerator and heat pump.

~ End of Unit II ~

THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – III : Second Law of Thermodynamics

★ Note: Syllabus: Kelvin-Planck & Clausius statements, equivalence, PMM-II, Carnot's principle & cycle, thermodynamic temperature scale, Clausius inequality, entropy, principle of entropy increase, availability & irreversibility, Gibbs & Helmholtz functions, Maxwell relations, third law

1. Need for the Second Law

The first law only balances energy. It cannot say which way a process occurs, nor how much heat can be turned into work. Observations:

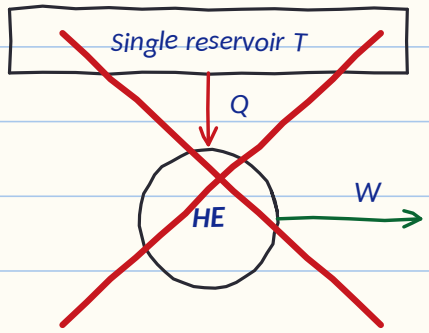
- Heat flows from hot to cold, never spontaneously from cold to hot.
- Work can be completely converted into heat (friction), but heat cannot be completely converted into work in a cycle.
- Gas expands into vacuum but never compresses itself back; mixing is never un-mixed by itself.

★ Note: The second law gives the direction of processes, the quality of energy, and the maximum possible efficiency of heat engines.

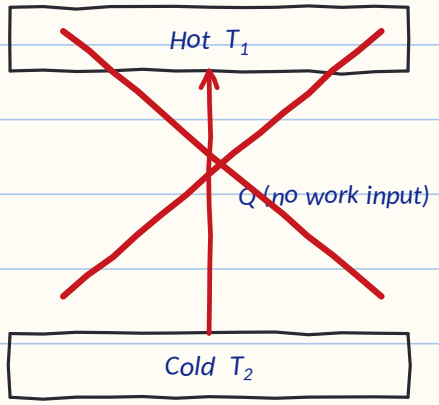
2. Statements of the Second Law

★ Note: Kelvin–Planck statement: It is impossible for a device operating in a cycle to receive heat from a single reservoir and produce a net amount of work.

★ *Note:* Clausius statement: It is impossible to construct a device operating in a cycle whose sole effect is the transfer of heat from a cooler body to a hotter body.



Kelvin–Planck violated

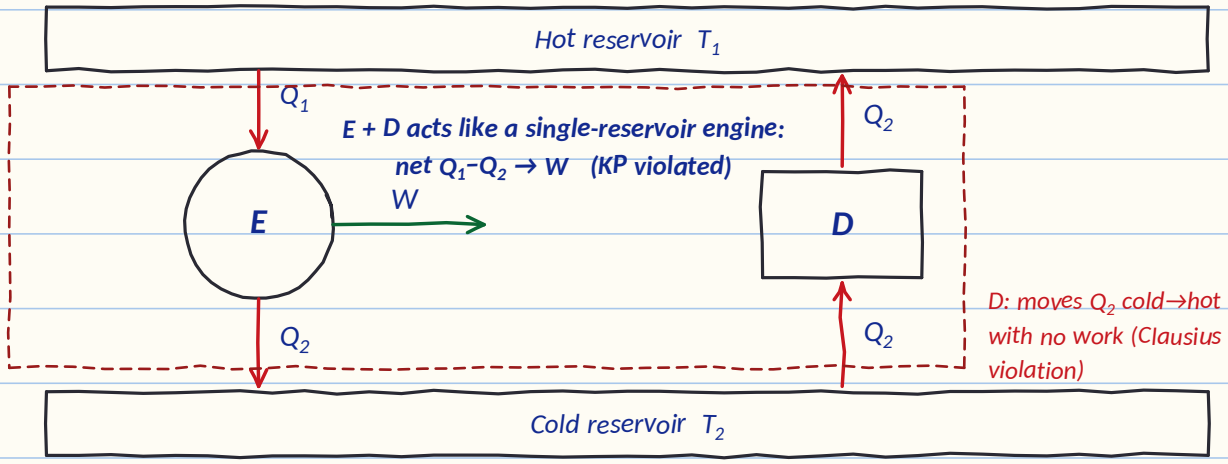


Clausius violated

- K-P: a heat engine must reject some heat to a sink ⇒ thermal efficiency is always less than 100 %.
- Clausius: a refrigerator needs work input; COP_R cannot be infinite.
- K-P deals with heat engines; Clausius with refrigerators / heat pumps. Both are negative statements – proved by the absence of any contrary experience.

3. Equivalence of Kelvin–Planck and Clausius Statements

The two statements are equivalent: violating one means violating the other.



(a) Violation of Clausius ⇒ violation of Kelvin–Planck

→ Let device D move Q_2 from cold (T_2) to hot (T_1) with no work (violates Clausius).

→ Engine E takes Q_1 from hot reservoir, produces W and rejects Q_2 to the cold reservoir.

→ Combined (E + D): the cold reservoir is unchanged; the hot reservoir loses $Q_1 - Q_2$ and a net work $W = Q_1 - Q_2$ is produced from a single reservoir ⇒ K-P violated.

(b) Violation of Kelvin–Planck ⇒ violation of Clausius

→ Let engine X take Q from hot reservoir and convert it fully to work $W = Q$ (violates K-P).

→ Use W to drive a refrigerator R: it takes Q_2 from cold and rejects $Q + Q_2$ to hot.

→ Net: hot reservoir gains Q_2, cold loses Q_2, with no work input ⇒ Clausius violated.

4. Perpetual Motion Machine of Second Kind (PMM-II)

A heat engine that takes heat from a single reservoir and converts it completely into work, thereby having $\eta = 100\ \%$.

- It does not violate the first law (energy is conserved) but violates the second law (Kelvin-Planck).
- Example idea: extracting heat from ocean water to run a ship – impossible.
- So: PMM-I violates the first law; PMM-II violates the second law; both impossible.

5. Carnot's Principle (Carnot Theorems)

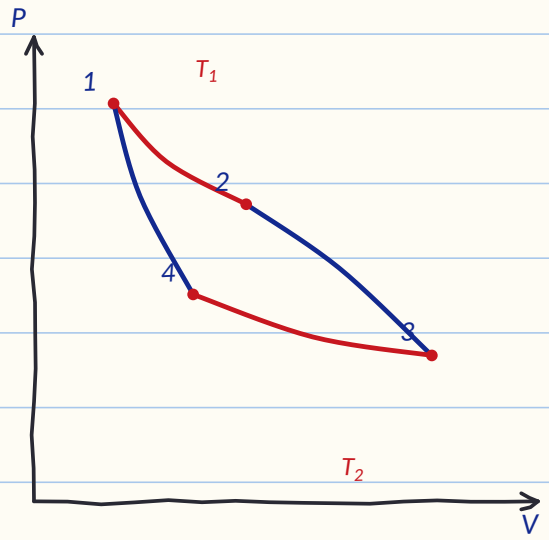
★ Note: Theorem 1: No heat engine operating between two given reservoirs can be more efficient than a reversible engine operating between the same two reservoirs.

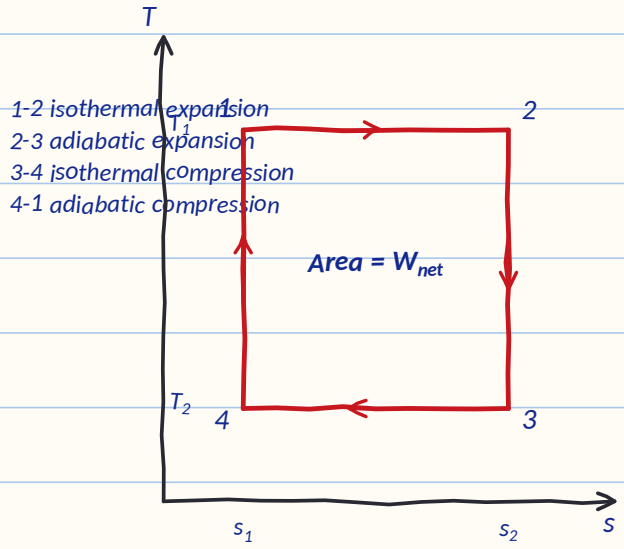
★ Note: Theorem 2: All reversible engines operating between the same two reservoirs have the same efficiency, whatever the working substance.

- Proof idea (Th.1): assume an irreversible engine has a higher efficiency; run a reversible engine as a heat pump with its work; combined device violates the Clausius / K-P statement.
- Hence $\eta_{rev} = f(T_1, T_2)$ only – leads to the thermodynamic temperature scale.

6. Carnot Cycle

A reversible cycle with two isothermal and two adiabatic (isentropic) processes, working between source T_1 and sink T_2.





Process	Nature	Heat / work
1-2	Isothermal expansion at T1	Q1 = mRT1 ln(V2/V1) absorbed
2-3	Adiabatic (isentropic) expansion	T falls T1→T2, work out
3-4	Isothermal compression at T2	Q2 = mRT2 ln(V3/V4) rejected
4-1	Adiabatic compression	T rises T2→T1, work in

From the adiabatic processes: $T_1/T_2 = (V_3/V_2)^{\gamma-1} = (V_4/V_1)^{\gamma-1} \Rightarrow V_2/V_1 = V_3/V_4$.

Hence the logarithmic terms cancel:

$$\eta_{Carnot} = W_{net}/Q_1 = 1 - Q_2/Q_1 = 1 - T_2/T_1$$

$$COP_{R,Carnot} = T_2/(T_1 - T_2) \qquad COP_{HP,Carnot} = T_1/(T_1 - T_2)$$

Specialties of the Carnot cycle

- It is completely reversible – the standard of perfection for all cycles.
- Its efficiency depends only on the reservoir temperatures, not on the working fluid.
- η increases when T_1 is raised or T_2 is lowered. η = 100 % only if T_2 = 0 K (impossible) or $T_1 = \infty$.
- The efficiency is the maximum possible between T_1 and T_2.
- Practical limits: needs infinitely slow isothermal heat transfer; very small work ratio; large V-ratio needed – hence not built in practice.

7. Thermodynamic (Kelvin) Scale of Temperature

Since η_{rev} depends only on temperatures, define the temperature ratio by the heat ratio of a reversible engine:

$$Q_1/Q_2 = T_1/T_2 \quad \text{(reversible engine)}$$

- Independent of any working substance or thermometric property – an *absolute* scale.
- Fixed point: triple point of water = 273.16 K. Then $T = 273.16\ (Q/Q_{tp})$.
- When $Q_2 \to 0$, $T_2 \to 0$: the absolute zero. Not attainable (third law).
- The ideal-gas scale coincides with the Kelvin scale.

8. Clausius Inequality

★ Note: For any cycle: $\oint \delta Q/T \le 0$. Equality holds for reversible cycles, strict inequality for irreversible cycles.

- Reversible cycle: $\oint \delta Q/T = 0$. Irreversible: $\oint \delta Q/T < 0$. Impossible cycle: $\oint \delta Q/T > 0$.
- Proof sketch: the cyclic system exchanges δQ at T with a Carnot engine linked to a reservoir T_0; apply K-P to the composite – net work cannot be positive $\Rightarrow \oint \delta Q/T \le 0$.
- Check for a cycle: $Q_1 = 1000$ kJ at 500 K, Q_2 rejected at 300 K. $\oint \delta Q/T = 2 - Q_2/300$. $Q_2 = 600 \Rightarrow 0$ (reversible); $700 \Rightarrow -0.33$ (irreversible, possible); $500 \Rightarrow +0.33$ (impossible).

9. Entropy

From $\oint (\delta Q/T)_{rev} = 0$, the quantity $\delta Q_{rev}/T$ is an exact differential – a property called entropy S.

$$dS = (\delta Q/T)_{rev} \qquad \Delta S = S_2 - S_1 = \int (\delta Q/T)_{rev}$$

- Unit kJ/K (extensive); specific entropy s in kJ/kg·K.
- Reversible adiabatic process: $dS = 0$ – isentropic. For irreversible processes: $dS > \delta Q/T$.

$$dS \geq \delta Q/T \quad \Rightarrow \quad \Delta S = \int \delta Q/T + S_{gen}, \quad S_{gen} \geq 0$$

Entropy change relations

$$\text{Ideal gas: } s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln(v_2/v_1)$$

$$s_2 - s_1 = c_p \ln(T_2/T_1) - R \ln(P_2/P_1)$$

- Isothermal heat transfer: $\Delta S = Q/T$. Phase change at T_{sat}: $\Delta s = h_{fg}/T_{sat}$. Liquid / solid: $\Delta s = c \ln(T_2/T_1)$.
- T-s diagram: area under a reversible curve = heat transfer $\int T\, ds$.

T ds relations

$$T\,ds = du + P\,dv \qquad T\,ds = dh - v\,dP$$

10. Principle of Entropy Increase

★ Note: The entropy of an isolated system (or of the universe) never decreases: $(\Delta S)_{isolated} \geq 0$; $\Delta S_{universe} = \Delta S_{system} + \Delta S_{surr} \geq 0$.

- > 0 irreversible process; $= 0$ reversible process; < 0 impossible.
- Entropy is not conserved; it is generated by irreversibilities (friction, finite ΔT heat transfer, mixing, free expansion).
- This is the quantitative form of the second law; gives the direction of natural processes.

Example: heat transfer through finite ΔT

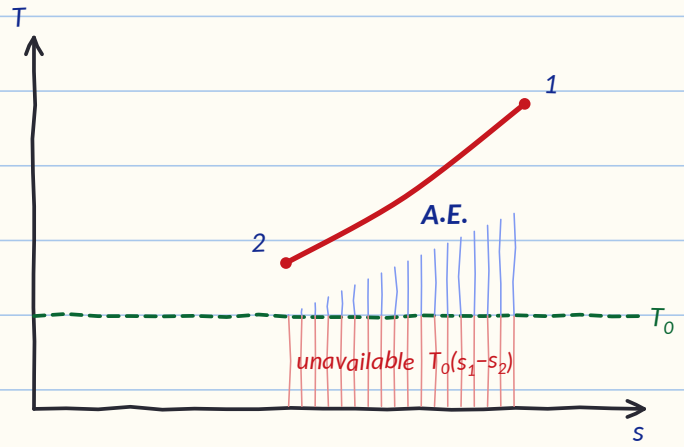
1000 kJ from 600 K body to 300 K body:

$\Delta S_{hot} = -1000/600 = -1.667$ kJ/K ; $\Delta S_{cold} = +1000/300 = +3.333$ kJ/K

$\Delta S_{univ} = +1.667$ kJ/K > 0 ⇒ irreversible

11. Availability and Irreversibility

Available energy (exergy): maximum useful work obtainable from a system as it comes to equilibrium with the environment (dead state: P_0, T_0). The rest is unavailable energy.



Available energy = $Q - T_0\Delta S$

Unavailable = $T_0\Delta S$

Max useful work from heat
→ between T and dead-state T_0

Heat from a constant-T source: $A.E. = Q\,(1 - T_0/T)$

Closed system: $\Phi = (U - U_0) + P_0(V - V_0) - T_0(S - S_0)$

Steady flow: $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + g\,z$

Irreversibility

$I = W_{rev} - W_{actual} = T_0\,S_{gen}$ $\qquad$ $\eta_{II} = W_{actual}/W_{rev}$

- Irreversibility = lost work (destroyed availability). Zero for a reversible process.
- Example: 1000 kJ at 600 K, T_0 = 300 K gives A.E. = 500 kJ. Transfer directly to a body at 400 K: $S_{gen} = 1000(1/400 - 1/600) = 0.833$ kJ/K ⇒ $I = 300 \times 0.833 = 250$ kJ lost.

- Second law efficiency compares actual performance to the best possible – it pinpoints where energy quality is wasted.

12. Thermodynamic Potentials – Helmholtz & Gibbs Functions

Helmholtz function $F = U - TS$

Gibbs function $G = H - TS$

Potential	Differential form	Natural variables
Internal energy U	$dU = T\,dS - P\,dV$	S, V
Enthalpy H	$dH = T\,dS + V\,dP$	S, P
Helmholtz F	$dF = -S\,dT - P\,dV$	T, V
Gibbs G	$dG = -S\,dT + V\,dP$	T, P

- Helmholtz: the maximum work obtainable from a closed system in a reversible isothermal process is $-\Delta F$ (at constant T, V: $dF \le 0$ for spontaneity).
- Gibbs: criterion of equilibrium at constant T and P: $dG \le 0$ spontaneous, $dG = 0$ equilibrium. During a phase change $g_f = g_g$.
- These are state functions (exact differentials), so cross-differentiation gives the Maxwell relations.

13. Maxwell Relations

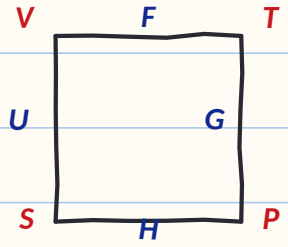
If $z = f(x, y)$ with $dz = M\,dx + N\,dy$, then $(\partial M/\partial y)_x = (\partial N/\partial x)_y$. Applying to the four potentials:

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V \quad \text{[from } dU\text{]}$$

$$\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P \quad \text{[from } dH\text{]}$$

$$(\partial S/\partial V)_T = (\partial P/\partial T)_V \qquad [\text{from } dF]$$

$$(\partial S/\partial P)_T = -(\partial V/\partial T)_P \qquad [\text{from } dG]$$



Edge letters = potentials: U, F, G, H

Corner letters = natural variables

Good Physicists Have Studied Under Very Fine Teachers (clockwise from top-left: V F T G P H S U)

- Use: replace hard-to-measure entropy derivatives by measurable P-V-T derivatives. Basis of the Clapeyron equation and of the Tds equations.

14. Third Law of Thermodynamics (Elementary)

★ Note: Nernst heat theorem / Third law: the entropy of a pure, perfectly crystalline substance is zero at absolute zero temperature.

- As $T \to 0$: $S \to 0$ and specific heats $\to 0$; this gives a reference for absolute entropy values.
- Absolute zero cannot be reached in a finite number of steps (unattainability principle).
- Heat capacity $c_p \to 0$ as $T \to 0$; thermal expansion coefficient $\to 0$.

15. Solved Numerical Problems

Problem 1 : Carnot efficiency

Engine between 800 K and 300 K: $\eta_{max} = 1 - 300/800 = 0.625 = 62.5\ \%$

Problem 2 : Carnot refrigerator

Cold space −10 °C (263 K), surroundings 27 °C (300 K):

$COP = 263/(300 - 263) = 7.11$

Problem 3 : entropy change of water

2 kg water heated 20 °C → 80 °C, $c = 4.18$ kJ/kg·K:

$\Delta S = m\, c\, \ln(T_2/T_1) = 8.36 \times \ln(353.15/293.15) = 1.56$ kJ/K

Problem 4 : isothermal expansion of air

1 kg air, volume doubles at constant T:

$\Delta S = m\, R\, \ln(V_2/V_1) = 0.287 \times \ln 2 = 0.199$ kJ/K

Problem 5 : Is the engine possible?

An engine receives 1000 kJ at 500 K and rejects 600 kJ at 300 K; claims to produce 400 kJ.

$\eta = 400/1000 = 40\ \%$ = Carnot limit $(1 - 300/500) \Rightarrow$ reversible, only just possible.

If it produced 500 kJ ($\eta = 50\ \%$) – impossible (> Carnot).

16. Entropy Changes in Ideal-Gas Processes

Process	Δs (per kg)
Isochoric (v const)	$c_v \ln(T2/T1) = c_v \ln(P2/P1)$
Isobaric (P const)	$c_p \ln(T2/T1) = c_p \ln(v2/v1)$
Isothermal	$R \ln(v2/v1) = -R \ln(P2/P1)$
Isentropic	0
Polytropic PV^n=C	$c_v(n-\gamma)/(n-1) \ln(T2/T1)$

- On the T-s diagram: isochoric lines are steeper than isobaric lines ($c_v < c_p$); isothermal = horizontal; isentropic = vertical.

17. Statistical Meaning of Entropy

- Boltzmann: $S = k \ln W$, where W = number of microstates for a given macrostate and $k = 1.38 \times 10^{-23}$ J/K.
- Entropy measures molecular disorder: ice → water → steam increases entropy; crystal at 0 K has $W = 1$, so $S = 0$ (third law).
- Natural processes move toward the most probable (most disordered) state; hence entropy of the universe increases – the "arrow of time".

18. First Law vs Second Law

First law	Second law
Conservation of energy	Quality and direction of energy
$Q - W = \Delta E$	$dS \geq \delta Q/T$
Defines internal energy U	Defines entropy S
Does not forbid PMM-II	Forbids PMM-II
Heat ⇌ work equivalence	Heat cannot be fully converted to work

Improving real-engine efficiency

→ Raise source temperature T_1 (limited by materials).

→ Lower sink temperature T_2 (limited by atmosphere / river water).

→ Reduce irreversibilities: friction, finite ΔT heat transfer, throttling, unrestrained expansion.

19. More Solved Problems

Problem 6 : free expansion

1 kg air, volume doubles by free expansion ($Q = 0$, $W = 0$):

$\Delta S_{sys} = 0.287 \ln 2 = 0.199$ kJ/K ; $\Delta S_{surr} = 0 \Rightarrow \Delta S_{univ} = +0.199$ kJ/K

Problem 7 : Carnot heat pump

Room 20 °C (293 K), outside 0 °C (273 K):

$COP_{HP} = 293/(293 - 273) = 14.65$

Problem 8 : Carnot engine

Output 30 kW between 600 K and 300 K: $\eta = 50\ \%$

$Q_1 = 30/0.5 = 60$ kW ; $Q_2 = 30$ kW

Problem 9 : melting ice

5 kg ice at 0 °C melts ($h_{if} = 334$ kJ/kg):

$\Delta S = m\, h_{if}/T = 5 \times 334/273.15 = 6.11$ kJ/K

Problem 10 : irreversibility of a real engine

Engine takes 100 kW at 900 K, produces 40 kW, rejects 60 kW at $T_0 = 300$ K.

$S_{gen} = 60/300 - 100/900 = 0.0889$ kW/K

$I = T_0 S_{gen} = 26.7$ kW ; $W_{rev} = 100(1 - 300/900) = 66.7$ kW

$\eta_{II} = 40/66.7 = 0.60$

20. Quick Revision & Important Questions

- K-P ⇒ no 100 % engine; Clausius ⇒ no free heat flow cold→hot; equivalent.
- $\eta_{Carnot} = 1 - T_2/T_1$; $\oint \delta Q/T \le 0$; $dS \ge \delta Q/T$; $\Delta S_{univ} \ge 0$.
- $F = U - TS$; $G = H - TS$; four Maxwell relations; $S(0\ K) = 0$.

→ State and prove the equivalence of Kelvin–Planck and Clausius statements.

→ Describe the Carnot cycle on P-V and T-s diagrams and derive its

efficiency.

→ State and prove the Clausius inequality. Define entropy.

→ What is availability? Derive irreversibility. Define second-law efficiency.

→ Derive Maxwell relations and explain Gibbs and Helmholtz functions.

~ End of Unit III ~

THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – IV : Pure Substances & Steam Properties

★ Note: Syllabus: Pure substances, P-V-T surface, T-s & h-s diagrams, Mollier charts, phase transformations, triple point & critical state, dryness fraction, Clausius-Clapeyron equation, property tables, thermodynamic processes of steam, steam calorimetry

1. Pure Substance

A pure substance has a homogeneous and invariable chemical composition throughout its mass, even if the phase changes.

- Examples: water (ice, water, steam), nitrogen, carbon dioxide, refrigerant R-134a, air (gas phase only, as a mixture of fixed composition).
- A mixture of water and steam is pure; a mixture of liquid air and gaseous air is NOT pure (composition differs in phases).
- Simple compressible substance: only P-V work; two independent properties fix the state.
- Gibbs phase rule: $F = C - P + 2$ (F = degrees of freedom, C = components, P = phases). Single-phase pure substance: $F = 2$; two-phase: $F = 1$; triple point: $F = 0$.

2. Phase Transformations of Water at Constant Pressure

Heat 1 kg of ice/water in a piston-cylinder at constant pressure and watch T and v:

→ Sub-cooled (compressed) liquid – $T < T_{sat}$; heating raises T. Sensible heat.

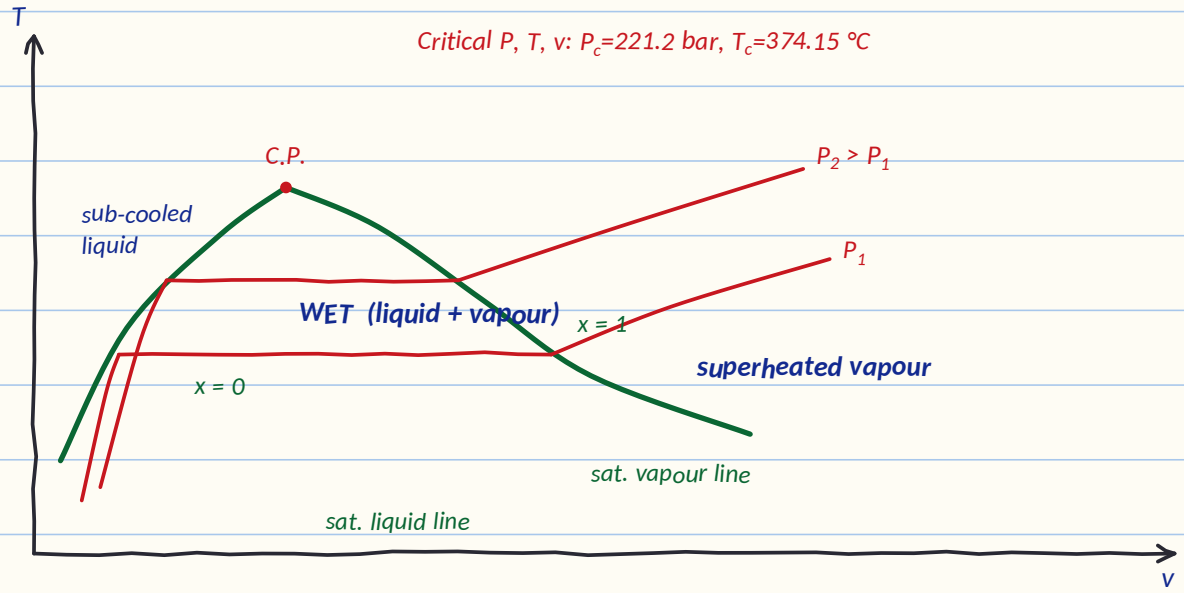
→ Saturated liquid – liquid at T_{sat} (about to evaporate). Properties with

subscript f.

→ Wet (saturated mixture) – liquid + vapour coexist at constant T_{sat}; heat added is latent heat; dryness fraction x rises 0 → 1.

→ Dry saturated vapour – x = 1; last drop evaporated. Subscript g.

→ Superheated vapour – $T > T_{sat}$; heating raises T again.

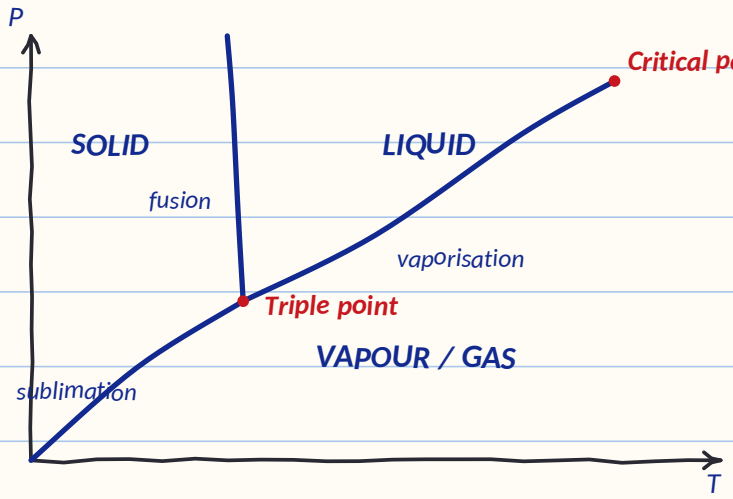


- As pressure increases, T_{sat} rises and the horizontal evaporation line (h_{fg}) becomes shorter; at the critical point it shrinks to a point.
- Degree of superheat = $T_{sup} - T_{sat}$. Degree of sub-cooling = $T_{sat} - T$.

3. Triple Point and Critical Point

- Triple point: the unique state at which solid, liquid and vapour phases coexist in equilibrium. Water: 0.01 °C, 0.00611 bar (0.6113 kPa).
- Critical point: the state at which the liquid and vapour phases become identical – no distinct evaporation, $h_{fg} = 0$. Above it, a fluid is "supercritical".

Water critical constants: P_c = 221.2 bar, T_c = 374.15 °C, v_c = 0.00317 m^3/kg



Triple point of water:
0.01 °C , 0.00611 bar

Critical point of water:
374.15 °C , 221.2 bar

Triple point is a fixed point
($F = C - P + 2 = 0$)

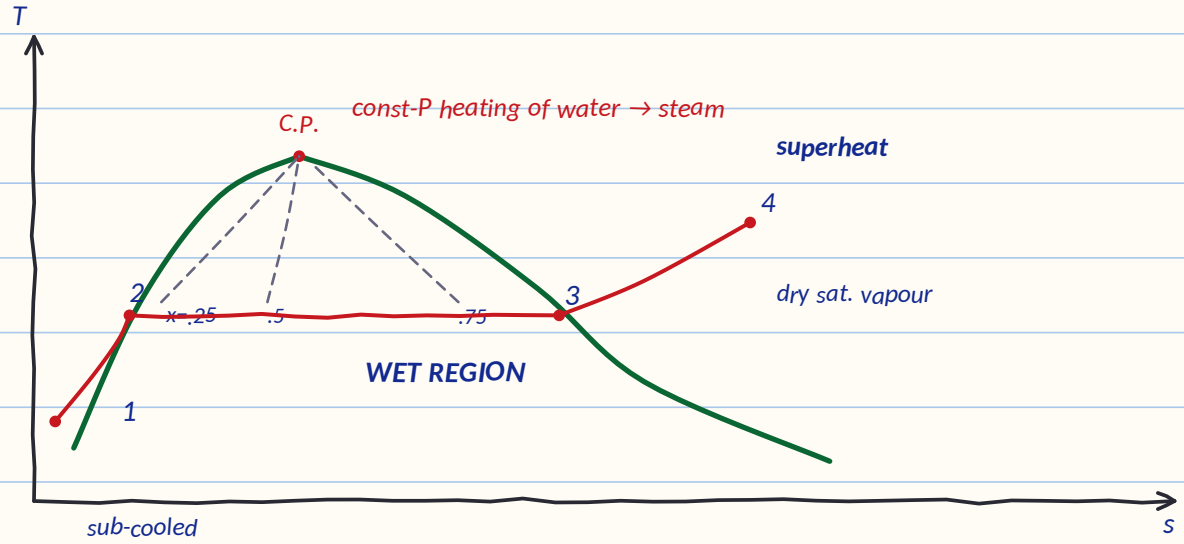
- P-T diagram: sublimation curve (solid ⇌ vapour), fusion curve (solid ⇌ liquid), vaporisation curve (liquid ⇌ vapour, ends at the critical point).
- For water the fusion curve slopes backward (ice expands on freezing, melting point falls as P rises); for most substances it slopes forward.

4. P-v-T Surface

The equilibrium states of a pure substance form a surface in three dimensions (P, v, T).

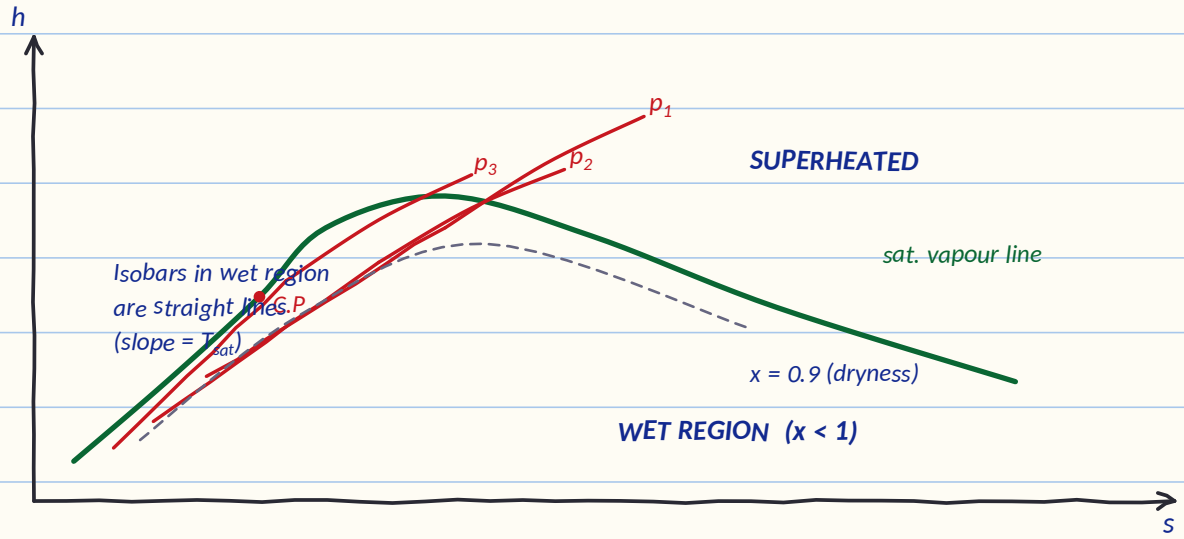
- Single-phase regions (solid, liquid, vapour) are curved surfaces; two-phase regions (solid-vapour, solid-liquid, liquid-vapour) are ruled surfaces parallel to the v-axis because P and T are fixed during phase change.
- The triple line is a line of constant P and T where three phases coexist; its projection on the P-T plane is the triple point.
- Projections: P-T diagram (phase boundaries), P-v diagram, T-v diagram – the dome shape of the liquid-vapour region is seen in the last two.
- Substances that contract on freezing (most) vs expand on freezing (water, antimony, bismuth) have slightly different surfaces.

5. T-s Diagram of Water



- Left of the critical point the saturated liquid line ($x = 0$); right the saturated vapour line ($x = 1$); inside the dome = wet region.
- Constant pressure heating 1→2→3→4: sub-cooled → saturated liquid → dry steam → superheated steam.
- Constant dryness lines ($x = 0.25, 0.5, 0.75$) converge at the critical point.
- Area under a reversible curve = heat transferred ($\int T\,ds$). Reversible adiabatic = vertical line.
- Latent heat of vaporisation $h_{fg} = T_{sat}\, s_{fg}$ (area under the horizontal line).

6. Mollier (h-s) Diagram



- Mollier chart plots enthalpy (ordinate) against entropy (abscissa). Isentropic process = vertical line; throttling = horizontal line (h const).
- Constant-pressure lines: straight lines with slope = T_{sat} in the wet region (they diverge upward), curves in the superheated region.
- Constant-dryness lines, constant-temperature lines (superheat region) and the saturation line are given.
- Advantage: enthalpy drop in turbines and nozzles read directly as a vertical distance. Practical working region is mostly wet and superheated steam.

7. Dryness Fraction and Properties of Steam

★ Note: Dryness fraction x = mass of dry vapour / total mass of mixture = $m_g/(m_f + m_g)$. Wetness fraction = 1 − x. Liquid: x = 0; dry saturated: x = 1.

$$v = v_f + x\, v_{fg} \approx x\, v_g$$

$$h = h_f + x\, h_{fg} \qquad u = u_f + x\, u_{fg}$$

$$s = s_f + x\, s_{fg} \qquad h_{fg} = h_g - h_f$$

- Superheated steam: $h = h_g + c_{ps}(T_{sup} - T_{sat})$, $c_{ps} \approx 2.1$ kJ/kg·K (approx.).
- Sub-cooled liquid: $h \approx h_f$ at the given T, $v \approx v_f$ (almost incompressible).
- Internal energy: $u = h - P\,v$. Enthalpy of vaporisation decreases with pressure and is 0 at the critical point.

Property table (short extract – saturated water)

P (bar)	T_s °C	h_f	h_{fg}	h_g	v_g m³/kg	s_g
1	99.6	417.5	2257.9	2675.4	1.694	7.359
10	179.9	762.6	2013.6	2776.2	0.1944	6.586

8. Property Tables

- Saturation table (temperature based): gives P_{sat}, v_f, v_g, u, h, s at a given T. Saturation table (pressure based): gives T_{sat} etc. at a given P.
- Superheated table: at a pressure, lists v, u, h, s for various temperatures above T_{sat}.
- Compressed-liquid table: used at high pressure, otherwise approximate with saturated liquid at the same T.
- Decide the region first: compare given v or h with v_f, v_g (or h_f, h_g). $v_f < v < v_g \Rightarrow$ wet, so use x.
- Use linear interpolation between table rows when the exact value is not tabulated.

9. Clausius–Clapeyron Equation

Gives the slope of the saturation (vaporisation) curve on a P-T diagram. Starting from the Maxwell relation $(\partial P/\partial T)_v = (\partial s/\partial v)_T$, in the two-phase region P depends on T only:

$$\frac{dP}{dT} = \frac{s_{fg}}{v_{fg}} = \frac{h_{fg}}{T\,v_{fg}} \qquad \text{(Clapeyron)}$$

- For liquid-vapour: v_f negligible, vapour ideal $\Rightarrow v_{fg} \approx RT/P$:

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{h_{fg}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \qquad \text{(Clausius–Clapeyron)}$$

- Applies to fusion ($dP/dT = h_{if}/(T\,v_{if})$, negative for water) and sublimation too.
- Example: at 100 °C, $h_{fg} = 2257$ kJ/kg, $v_{fg} = 1.672$ m³/kg $\Rightarrow dP/dT = 2257\times10^3/(373.15 \times 1.672) = 3.62$ kPa/K.

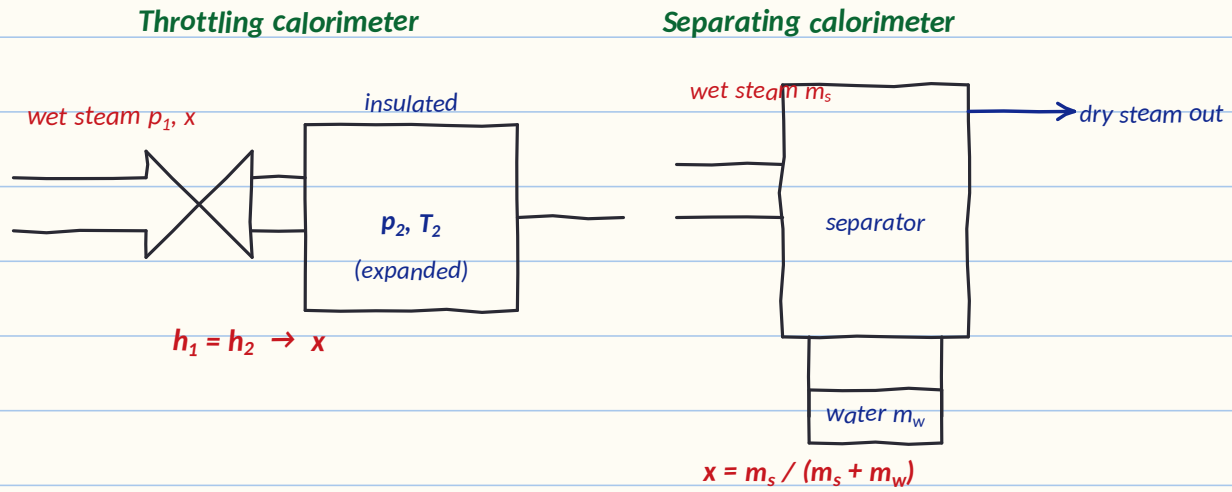
10. Thermodynamic Processes of Steam

Process	Work W	Heat Q	Remarks
Constant volume	0	u2 − u1	v1 = v2, find x2 or T2
Constant pressure	P(v2−v1)	h2 − h1	boiler, condenser
Isothermal	Q − Δu	T(s2 − s1)	wet: also const P
Isentropic (rev. adiabatic)	u1 − u2	0	turbine, nozzle; s1 = s2
Throttling	0	0	h1 = h2, irreversible
Polytropic P v^n = C	(P1v1−P2v2)/(n−1)	Δu + W	n=1.3 superheat, 1.135 wet

- Isentropic expansion in turbine: from state 1 draw a vertical line on h-s chart to P_2; ideal work $w = h_1 - h_2$. Actual work smaller; isentropic efficiency $\eta = (h_1 - h_{2a})/(h_1 - h_{2s})$.
- Throttling: steam flows through a restriction, P drops, h constant, entropy increases, T may fall or the steam becomes superheated.

11. Steam Calorimetry

Measures the dryness fraction x of wet steam in a pipe / boiler.



Throttling calorimeter

- Wet steam of pressure P_1 is throttled (h const) to a lower pressure P_2 so that it becomes slightly superheated; T_2, P_2 are measured.

$$h_1 = h_2 \Rightarrow h_{f1} + x\, h_{fg1} = h_{g2} + c_{ps}(T_2 - T_{sat2})$$

$$x = [h_{g2} + c_{ps}(T_2 - T_{sat2}) - h_{f1}] / h_{fg1}$$

- Limitation: steam must be fairly dry (x > about 0.95) so that it is superheated after throttling; otherwise use a separating calorimeter first.

Separating calorimeter

- Steam sample is passed through a separator; the moisture is collected and weighed; remaining dry steam is condensed and weighed.

$$x = m_s/(m_s + m_w) \quad (m_s\text{: dry steam, } m_w\text{: separated water})$$

- Limitation: separates only part of the moisture – inaccurate for very dry steam.

Combined (separating + throttling) calorimeter

- Separating calorimeter first removes most moisture (x_1); the throttling part finds the dryness x_2 of what remains. Overall dryness $x = x_1\, x_2$.

12. Solved Numerical Problems

Problem 1 : properties of wet steam

Steam at 10 bar, $x = 0.9$. Find v, h, s, u.

$$v = 0.9 \times 0.1944 \approx 0.175\ m^3/kg\ ;\ h = 762.6 + 0.9 \times 2013.6 = 2574.8\ kJ/kg$$

$$s = 2.138 + 0.9 \times 4.448 = 6.141\ kJ/kg\cdot K$$

$$u = h - Pv = 2574.8 - 1000 \times 0.175 = 2400\ kJ/kg$$

Problem 2 : dryness fraction from volume

A rigid vessel of 0.3 m^3 holds 2 kg wet steam at 10 bar. Find x.

$$v = 0.3/2 = 0.15\ m^3/kg\ ;\ x = (v - v_f)/(v_g - v_f) = (0.15 - 0.001127)/(0.1944 - 0.001127) = 0.77$$

Problem 3 : heat to make steam

Heat required to convert 1 kg water at 30 °C to dry saturated steam at 1 bar ($h_{f30} = 125.7$ kJ/kg).

$$q = h_g - h_{f30} = 2675.4 - 125.7 = 2549.7\ kJ/kg$$

Problem 4 : superheated steam enthalpy

10 bar, 250 °C ($T_{sat} = 179.9$ °C): $h \approx 2776.2 + 2.1 \times (250 - 179.9) = 2923$ kJ/kg (approx.)

Problem 5 : isentropic turbine

Dry saturated steam at 10 bar expands isentropically to 1 bar. Find work.

$$s_2 = s_1 = 6.586 \Rightarrow x_2 = (6.586 - 1.3026)/6.0568 = 0.872$$

$$h_2 = 417.5 + 0.872 \times 2257.9 = 2387\ kJ/kg$$

$$w = h_1 - h_2 = 2776.2 - 2387 = 389\ kJ/kg$$

Problem 6 : throttling calorimeter

Steam at 10 bar is throttled to 1 bar, 110 °C ($h_2 \approx 2696$ kJ/kg). Find x.

$$x = (2696 - 762.6)/2013.6 = 0.96$$

Problem 7 : separating calorimeter

2 kg sample, 0.3 kg water separated ⇒ dry steam 1.7 kg ⇒ $x = 1.7/2 = 0.85$

13. Heat Added at Each Stage of Steam Formation

Stage	Heat per kg	Remarks
Water 0 °C → T_s	$h_f = c_{pw}T_s \approx 4.187\,T_s$	sensible heat
Evaporation at T_s	$x\,h_{fg}$	latent heat (wet steam)
Complete evaporation	h_{fg}	dry saturated steam, $h_g = h_f + h_{fg}$
Superheating	$c_{ps}(T_{sup} - T_s)$	$h_{sup} = h_g + c_{ps}\Delta T$

- Latent heat is spent in two ways: external work of evaporation $P(v_g - v_f)$ to push back the atmosphere, and internal latent heat = $h_{fg} - P(v_g - v_f)$ to separate the molecules.

  At 1 bar: $P\,v_{fg} = 100 \times (1.694 - 0.001) = 169.3$ kJ/kg

  Internal latent heat = 2257.9 − 169.3 = 2088.6 kJ/kg

- Total heat of steam from water at 0 °C: wet $h_f + x\,h_{fg}$; dry h_g; superheated $h_g + c_{ps}(T_{sup} - T_s)$.
- Specific volume: $v \approx x\,v_g$ (v_f negligible except near the critical point); superheated steam behaves approximately like a gas: $P\,v = 0.4615\,T$.

14. Real Gas Behaviour (Link with Pure Substances)

- At low P and high T a gas follows the ideal gas law $Pv = RT$. Near the dome it departs strongly.
- Compressibility factor $Z = P\,v/(R\,T)$. $Z = 1$ for ideal gas; $Z < 1$ where attraction dominates; $Z > 1$ at very high pressure.
- van der Waals equation: $(P + a/v^2)(v - b) = R\,T$; a corrects for intermolecular attraction, b for finite molecular volume.

$$a = 27\,R^2\,T_c^{\,2}/(64\,P_c) \qquad b = R\,T_c/(8\,P_c) \qquad Z_c = 3/8 = 0.375$$

- *Law of corresponding states*: all gases at the same reduced $P_r = P/P_c$ and $T_r = T/T_c$ have nearly the same $Z \Rightarrow$ generalized compressibility chart.
- Other equations: Redlich–Kwong, Beattie–Bridgeman, virial expansion $Z = 1 + B/v + C/v^2 + \ldots$

15. More Solved Problems

Problem 8 : constant pressure heating of wet steam

1 kg steam at 5 bar, $x = 0.8$ is heated at constant pressure until dry saturated ($h_{fg} = 2107.4$, $v_g = 0.3749\ m^3/kg$).

$$Q = (1 - 0.8) \times 2107.4 = 421.5\ kJ$$

$$v_1 \approx 0.8 \times 0.3749 = 0.300\ ; \quad W = P(v_g - v_1) = 500 \times 0.075 = 37.5\ kJ$$

$$\Delta u = Q - W = 421.5 - 37.5 = 384\ kJ$$

Problem 9 : external work of evaporation

At 10 bar: $P(v_g - v_f) = 1000 \times (0.1944 - 0.001127) = 193.3\ kJ/kg$

Internal latent heat $= 2013.6 - 193.3 = 1820.3\ kJ/kg$

Problem 10 : throttling process

Steam throttled from 10 bar to 1 bar, h constant. If $x_1 = 0.96$, $h_1 = 762.6 + 0.96 \times 2013.6 = 2695.9\ kJ/kg$.

h_g at 1 bar $= 2675.4 < 2695.9 \Rightarrow$ steam leaves superheated ; $T_2 \approx 100 + (2695.9 - 2675.4)/2.0 \approx 110\ °C$

Problem 11 : turbine isentropic efficiency

Ideal $\Delta h = 389\ kJ/kg$ (Problem 5); actual $\Delta h = 330\ kJ/kg$

$$\eta_{isen} = 330/389 = 0.85 = 85\ \%$$

16. Glossary & Short Questions

→ *Saturation temperature* – temperature at which a liquid boils at a given pressure; rises with P.

→ *Sensible heat* – heat that changes temperature; *latent heat* – heat that changes phase at constant T and P.

→ *Why is the critical point special?* Liquid and vapour properties coincide; no latent heat; beyond it the fluid is supercritical.

→ *Why do isobars in the wet region on T-s become horizontal?* P and T are both constant during phase change.

→ *Why does steam leave a throttle valve at lower temperature?* Enthalpy constant, pressure falls – saturation temperature drops, steam may become superheated.

→ *Why use the combined calorimeter?* Throttling alone fails for very wet steam; separating alone misses fine moisture.

17. Quick Revision & Important Questions

- Pure substance: fixed composition. Triple point F = 0. Critical point: h_{fg} = 0.
- $h = h_f + x\, h_{fg}$; $v = x\, v_g$; $s = s_f + x\, s_{fg}$. Throttling: h const. Isentropic: s const.
- $dP/dT = h_{fg}/(T\, v_{fg})$. Isobars in wet region are straight on T-s and h-s.

→ Draw the P-T and T-v diagrams of water and mark triple and critical points.

→ Explain the formation of steam at constant pressure with a T-s diagram.

→ Derive the Clausius–Clapeyron equation.

→ Describe the throttling and separating calorimeters. Derive the formula for x.

→ Explain the Mollier chart and its advantages.

~ End of Unit IV ~

THERMODYNAMICS

Course Code : 23A03301 | Handwritten Study Notes

UNIT – V : Refrigeration & Air Conditioning

★ Note: Syllabus: Introduction to refrigeration: air & vapour-compression systems, components, COP, refrigerants. Air conditioning: psychrometric properties & processes, sensible & latent heat loads, SHF, human comfort, effective temperature, comfort chart, load calculations

1. Introduction to Refrigeration

Refrigeration is the process of removing heat from a space or substance to bring it to a temperature lower than its surroundings and to maintain it there. It needs a work (or heat) input and a working fluid called the refrigerant.

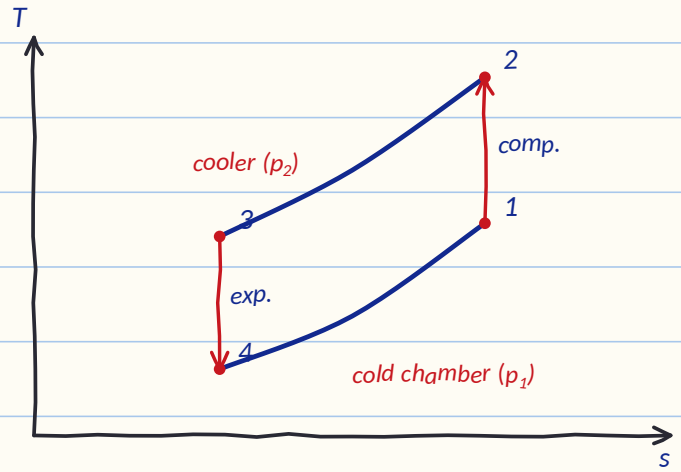
- Applications: food preservation, cold storage, ice plants, air conditioning, chemical industries, medical (vaccines, blood banks), cryogenics.
- Refrigerating effect (RE): heat extracted from the cold space per kg of refrigerant, kJ/kg. Capacity: kW or ton of refrigeration (1 TR = 3.517 kW = 211 kJ/min).

$$\text{COP} = \text{Refrigerating effect} / \text{Work input} = Q_2/W$$

- Reversed Carnot cycle (ideal limit): $\text{COP}_{max} = T_2/(T_1 - T_2)$, T_2 = evaporator temperature, T_1 = condenser temperature (K).
- Methods: vapour compression, vapour absorption, air refrigeration (gas cycle), thermoelectric, steam-jet.

2. Air Refrigeration – Bell–Coleman Cycle

Air is the working fluid (no phase change). It is the reversed Brayton (Joule) cycle, used in aircraft cooling and in cryogenic plants.



Air refrigeration (Bell–Coleman)

Reversed Brayton cycle

$RE = c_p(T_1 - T_4)$

$W = c_p[(T_2-T_1)-(T_3-T_4)]$

$COP = T_4 / (T_3 - T_4)$

$= 1/(r_p^{(\gamma-1)/\gamma} - 1)$

→ 1-2 Isentropic compression in compressor: $P_1 \rightarrow P_2$, $T_1 \rightarrow T_2$.

→ 2-3 Constant-pressure cooling in the cooler (heat Q_1 rejected to surroundings).

→ 3-4 Isentropic expansion in the expander: T falls to T_4 (lowest temperature).

→ 4-1 Constant-pressure heating in the cold chamber (refrigerating effect, Q_2 absorbed).

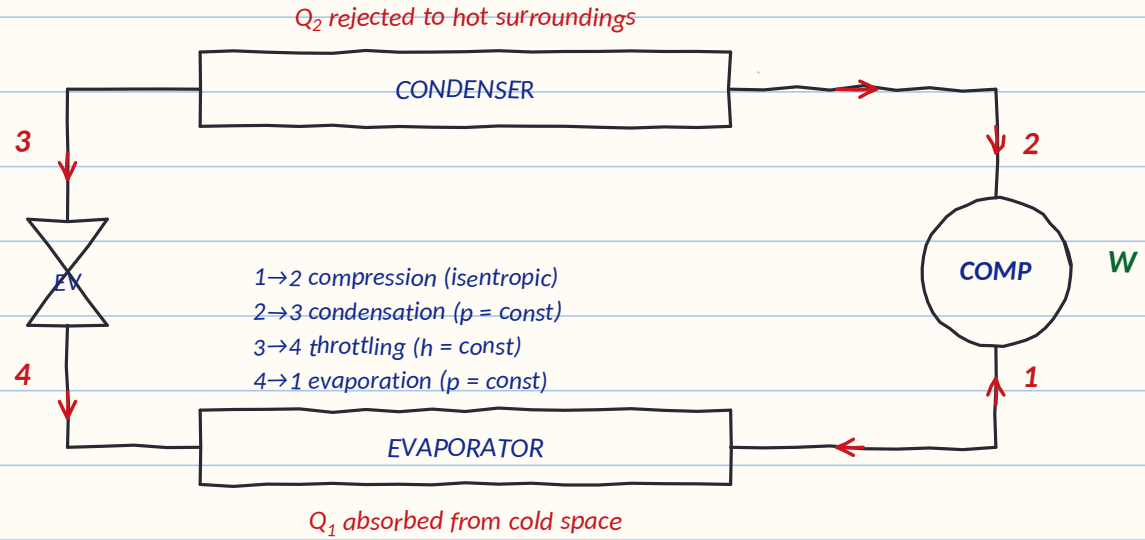
$$RE = c_p(T_1 - T_4) \qquad W = c_p[(T_2 - T_1) - (T_3 - T_4)]$$

$$COP = (T_1 - T_4) / [(T_2 - T_1) - (T_3 - T_4)] = 1/(r_p^{(\gamma-1)/\gamma} - 1)$$

- Advantages: air is cheap, non-toxic, non-flammable; light in weight – used on aircraft (cabin pressurised air is already available).
- Disadvantages: very low COP (< 1 for high r_p); large volume of air required; sensible-heat only, no latent heat; bulky equipment.
- Since the heat exchange is at varying temperature (not isothermal), COP is always lower than the Carnot COP between the same extremes.

3. Vapour Compression Refrigeration (VCR) System

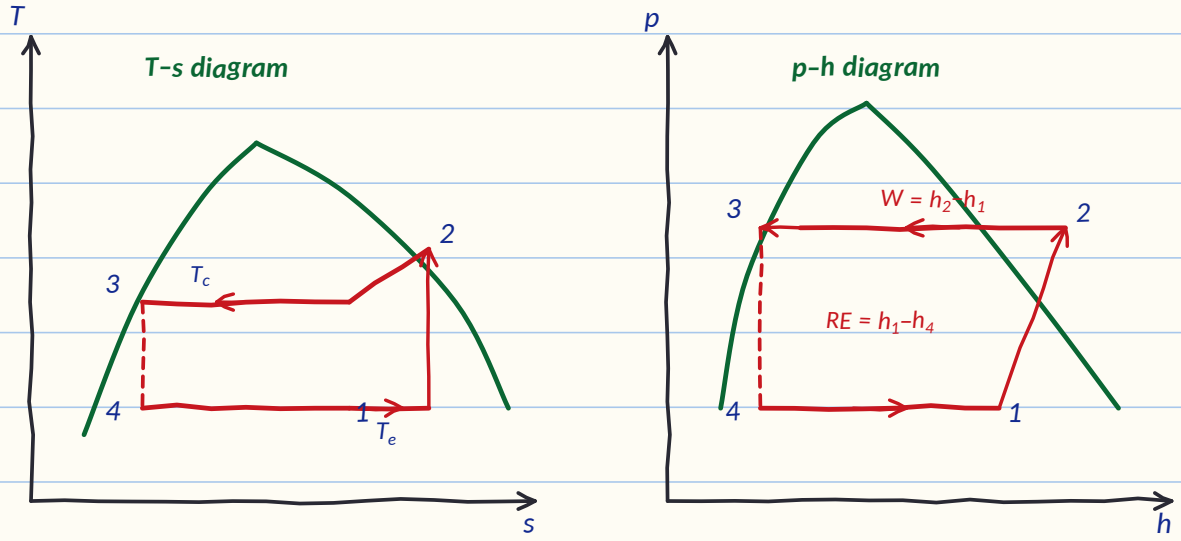
The most widely used system – the refrigerant undergoes phase change (evaporates, condenses), using the large latent heat.



Components

Component	Function
Compressor	Raises P and T of vapour (reciprocating, rotary, screw, centrifugal, scroll)
Condenser	Rejects heat; vapour → liquid at high P (air-/water-cooled)
Expansion valve	Throttles liquid to low P; T falls (capillary tube, TXV)
Evaporator	Liquid absorbs heat from the cold space and evaporates

Processes (T-s and p-h diagrams)



→ 1-2 Isentropic compression of dry saturated vapour to superheated vapour ($W = h_2 - h_1$).

→ 2-3 Constant-pressure cooling and condensation to saturated liquid in the condenser ($Q_1 = h_2 - h_3$).

→ 3-4 Throttling (isenthalpic): $h_3 = h_4$, irreversible; part of the liquid flashes to vapour.

→ 4-1 Constant-pressure, constant-temperature evaporation ($RE = h_1 - h_4$).

$$RE = h_1 - h_4 = h_1 - h_3 \qquad W = h_2 - h_1$$

$$COP = (h_1 - h_4) / (h_2 - h_1) \qquad Q_{cond} = h_2 - h_3 = RE + W$$

- Why a throttle valve instead of an expander? The work recovered from a small liquid-vapour expansion is very small, and the valve is simple, cheap and controllable.
- Dry compression (vapour enters dry) is preferred to wet compression (liquid droplets can damage the compressor).
- Sub-cooling the liquid before the valve and superheating the vapour before compression increase RE; superheat adds some work.
- Vapour compression advantages over air refrigeration: high COP,

compact, uses latent heat, nearly constant evaporator temperature.

Effect of operating conditions

- Lower evaporator pressure ⇒ lower RE and more work per kg ⇒ COP decreases. Higher condenser pressure ⇒ more work, less RE ⇒ COP decreases.
- Practical (actual) cycle deviates: pressure drops in pipes, heat gains, non-isentropic compression ($\eta_{isen} < 1$), sub-cooling and superheating.

4. Refrigerants

A refrigerant is the working fluid that absorbs heat at low T in the evaporator and rejects it at high T in the condenser.

Designation (ASHRAE R-number)

- R-xyz: x = C − 1, y = H + 1, z = F. R-12 = CCl_2F_2, R-22 = $CHClF_2$, R-134a = $CH_2F–CF_3$ (a = isomer).
- Inorganic: 700 + molecular mass: R-717 = ammonia (NH_3), R-718 = water, R-744 = CO_2. Hydrocarbons: R-290 propane, R-600a isobutane.

Desirable properties

- Thermodynamic: high latent heat, low freezing point, high critical temperature, moderate pressures (above atmosphere in evaporator), low specific volume of vapour, high COP.
- Physical / chemical: non-toxic, non-flammable, non-corrosive, chemically stable, miscible with lubricating oil, high thermal conductivity, easy leak detection, low cost.
- Environmental: zero ODP (ozone depletion potential) and low GWP (global warming potential).

Refrigerant	Type	Remarks
R-12 (CFC)	synthetic	high ODP – banned (Montreal Protocol)
R-22 (HCFC)	synthetic	low ODP, being phased out
R-134a (HFC)	synthetic	ODP 0, GWP ≈ 1430; domestic refrigerators
R-717 (NH_3)	natural	excellent COP, toxic; industrial
R-744 (CO_2)	natural	non-toxic, very high pressure
R-290 / R-600a	natural	flammable, low GWP
R-1234yf (HFO)	synthetic	GWP < 1; automotive A/C

- *Secondary refrigerants* (brine – NaCl, $CaCl_2$; glycol; chilled water) carry cold from the evaporator to the load; they do not change phase.

5. Introduction to Air Conditioning

Air conditioning is the simultaneous control of air temperature, humidity, cleanliness and motion in a space to suit the occupants or a process.

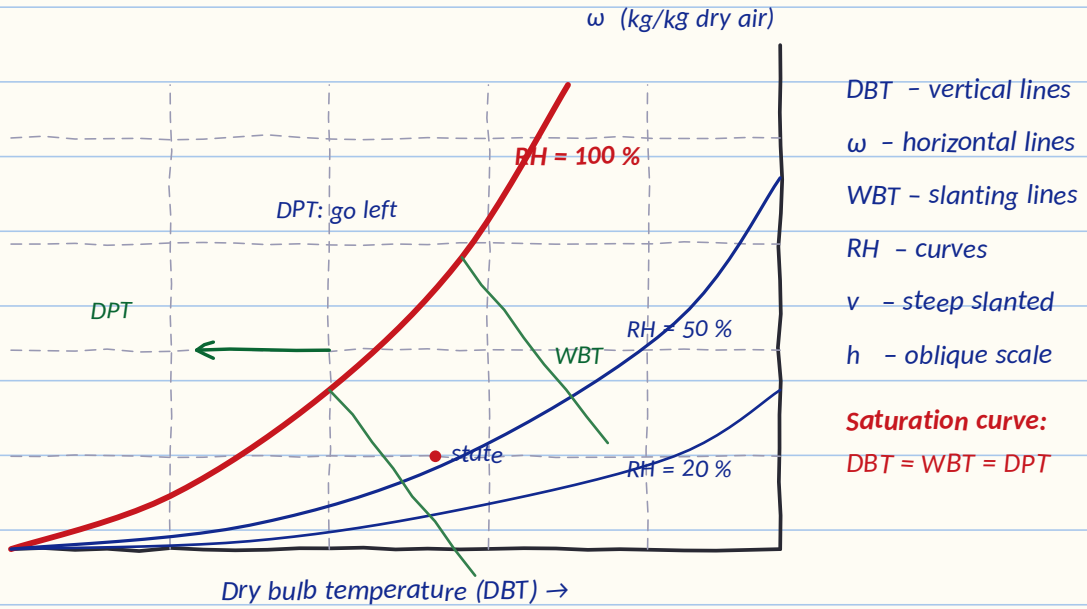
- Comfort air conditioning – for human comfort (homes, offices, theatres). Industrial A/C – for process requirements (textile, printing, pharma, electronics).
- By season: *summer* (cooling + dehumidification), *winter* (heating + humidification), *year-round*.
- Equipment: filter, cooling coil, heating coil, humidifier, fan / blower, ducts, diffusers, thermostat, refrigeration plant.

6. Psychrometric Properties

Moist air = dry air + water vapour. By Dalton's law the total pressure $P = p_a + p_v$.

Property	Definition / formula
Dry bulb temp (DBT) t	temperature shown by ordinary thermometer
Wet bulb temp (WBT) t_w	thermometer with wetted wick; evaporative cooling
Dew point temp (DPT)	temperature at which vapour starts condensing at constant P
Relative humidity φ	p_v/p_g at the DBT (p_g = saturation pressure)
Humidity ratio ω	$m_v/m_a = 0.622\ p_v/(P - p_v)$ kg/kg dry air
Degree of saturation μ	ω/ω_s at same DBT
Specific enthalpy h	$1.005\ t + \omega(2500 + 1.88\ t)$ kJ/kg dry air
Specific volume v	m^3 per kg dry air = R_aT/p_a

- DBT ≥ WBT ≥ DPT; for saturated air all three are equal and φ = 100 %.
- Carrier's equation (from a sling psychrometer): $p_v = p_w - (P - p_w)(t - t_w)/(1544 - 1.44\ t_w)$.
- Humid specific heat $c_{pm} = 1.005 + 1.88\ \omega \approx 1.022$ kJ/kg·K.



- The psychrometric chart shows these properties together: given any two independent properties, the rest are read from the chart.

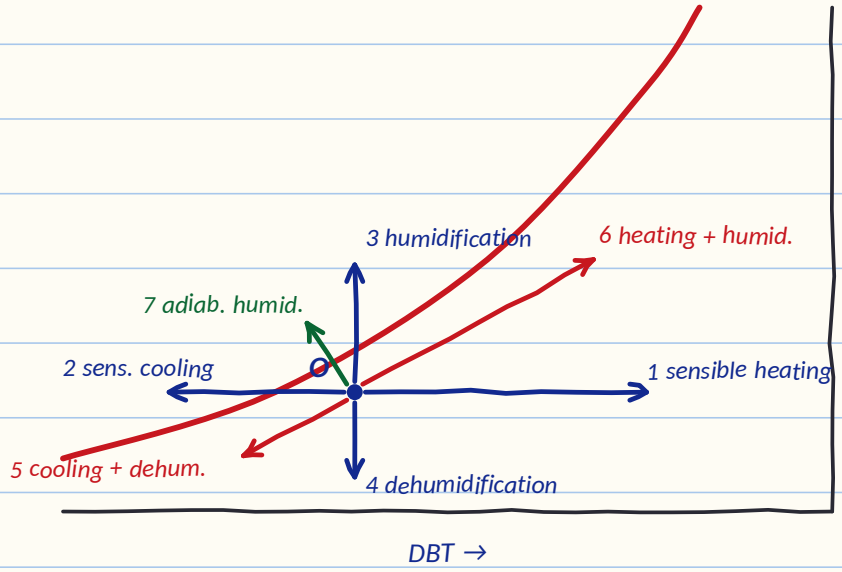
Example

$P = 1.01325$ bar, $p_v = 0.02$ bar, $t = 30$ °C ($p_g = 0.04246$ bar)

$\omega = 0.622 \times 0.02/(1.01325 - 0.02) = 0.0125$ kg/kg ; $\varphi = 0.02/0.04246 = 47$ %

$h = 1.005 \times 30 + 0.0125(2500 + 1.88 \times 30) = 62.1$ kJ/kg dry air

7. Psychrometric Processes



Sensible heat:

$Q_s = m\, c_{pm}(t_2 - t_1)$

Latent heat:

$Q_l = m\, h_{fg}(\omega_1 - \omega_2)$

SHF = $Q_s/(Q_s+Q_l)$

RSHF, GSHF, ESHF

Process	Path on chart	Example
Sensible heating	horizontal →, ω const	heating coil
Sensible cooling	horizontal ←, ω const (t > DPT)	dry cooling coil
Humidification	vertical ↑, t const	steam injection
Dehumidification	vertical ↓, t const	chemical absorbent
Cooling + dehumid.	down-left ↙	summer A/C coil
Heating + humid.	up-right ↗	winter A/C
Adiabatic humid.	along WBT line ↖	air washer

Adiabatic mixing of two air streams

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = (\dot{m}_1 + \dot{m}_2) h_3$$

$$\dot{m}_1 \omega_1 + \dot{m}_2 \omega_2 = (\dot{m}_1 + \dot{m}_2) \omega_3$$

$$t_3 \approx (\dot{m}_1 t_1 + \dot{m}_2 t_2)/(\dot{m}_1 + \dot{m}_2)$$

On the chart the mixed state lies on the straight line joining the two states, dividing it in the inverse ratio of the masses.

2 kg/s at 30 °C, ω = 0.020 with 1 kg/s at 20 °C, ω = 0.008:

$t_3 = (60 + 20)/3 = 26.7$ °C ; $\omega_3 = (0.040 + 0.008)/3 = 0.016$

Cooling coil – bypass factor

$$BF = (t_3 - t_{ADP})/(t_1 - t_{ADP}) \qquad \text{Contact factor} = 1 - BF$$

- ADP (apparatus dew point) = effective surface temperature of the coil. Smaller BF ⇒ better coil (more rows, lower air velocity).

8. Sensible and Latent Heat Loads – SHF

- Sensible heat changes DBT without changing moisture. Latent heat is associated with change in humidity ratio (moisture addition/removal).

$$Q_s = \dot{m}_a c_{pm} \Delta t = 1.23 \dot{V} \Delta t \quad (kW, \dot{V} \text{ in } m^3/s)$$

$$Q_l = \dot{m}_a \Delta\omega h_{fg} \approx 3000 \dot{V} \Delta\omega \quad (kW)$$

★ Note: Sensible Heat Factor SHF = $Q_s/(Q_s + Q_l) = Q_s/Q_{total}$. A low SHF means a large latent load (humid climate, crowded halls).

- RSHF = room SHF; GSHF = grand SHF including fresh-air load on the coil; ESHF = effective SHF taking coil bypass into account.
- SHF line on the chart: drawn from the room state with the slope given by the protractor; it fixes the supply air state and the ADP.

Example: $\dot{V} = 2\ m^3/s$, $\Delta t = 10\ K$, $\Delta\omega = 0.0015$

$Q_s = 1.23 \times 2 \times 10 = 24.6\ kW$; $Q_l = 3000 \times 2 \times 0.0015 = 9\ kW$

$SHF = 24.6/33.6 = 0.73$

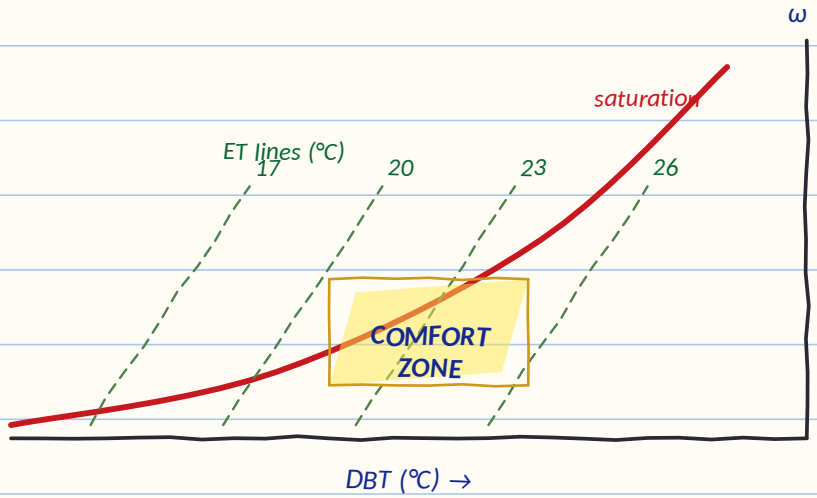
9. Human Comfort

The human body produces about 100 W of heat at rest (metabolism) and must lose it to keep the core near 37 °C. Heat loss: convection, radiation, evaporation (skin and breath), small conduction.

- Comfort is the condition of mind that expresses satisfaction with the thermal environment (ASHRAE).
- Factors: dry bulb temperature, humidity, air velocity, mean radiant temperature, activity level, clothing, age, sex, and cleanliness (odour, dust, CO_2).
- High humidity restricts sweat evaporation (hot-humid feels worse); low humidity dries the skin; air motion raises evaporation and convection.
- Recommended: DBT 22–27 °C, RH 40–60 %, air velocity 5–8 m/min, fresh air ≈ 7–10 L/s per person.

10. Effective Temperature and Comfort Chart

★ Note: Effective Temperature (ET) is an empirical index combining DBT, humidity and air velocity: it is the temperature of still, saturated air that gives the same sensation of warmth or cold.



Effective temperature (ET):
index combining DBT, humidity & air velocity → same feel

Typical comfort zone:
DBT 22–27 °C, RH 30–70 %
Air velocity ≈ 5–8 m/min

- Developed by Houghten and Yaglou (ASHVE). The comfort chart has DBT on the abscissa, WBT on the oblique scale, with ET lines and a shaded comfort zone.
- Persons at the same ET feel the same, though DBT and RH differ. Comfort zone ET ≈ 21–25 °C (approx.); winter zone is slightly lower.
- Limitations: overestimates humidity effect at low temperatures; ignores radiation – replaced by the corrected effective temperature (globe temperature) and PMV-PPD indices.

11. Comfort Air Conditioning and Load Calculations

- Summer: air is cooled and dehumidified by a cooling coil (process ↙), then reheated slightly if required, and supplied to the room.
- Winter: air is preheated, humidified (steam / spray) and heated to the supply state (process ↗).

Cooling load components

Sensible loads	Latent loads
Heat conduction through walls, roof, glass	Moisture from occupants (sweat, breath)
Solar radiation through windows	Moisture in infiltration & ventilation air
People (≈ 70–75 W sensible each)	Cooking, steam, open water surfaces
Lights, motors, appliances	Moisture from equipment / process
Infiltration and ventilation air	People (≈ 45–55 W latent each)

- Steps: (1) fix inside & outside design conditions; (2) compute envelope gains $Q = U\,A\,\Delta T_{eq}$; (3) add internal gains; (4) add ventilation & infiltration loads; (5) add safety factor and duct gains (5–10 %); (6) find SHF, supply air quantity, coil capacity (TR).

$$\text{Supply air: } \dot{V} = Q_{s,room} / [1.23\,(t_{room} - t_{supply})] \quad (m^3/s)$$

- Winter heating load is mainly conduction loss $Q = U\,A\,(t_i - t_o)$ plus ventilation/infiltration load.

Solved load example

Office, 20 persons: people $20 \times 75 = 1.5$ kW sensible, $20 \times 55 = 1.1$ kW latent

Lights 2 kW, equipment 1.5 kW, solar + transmission 6 kW

Sensible = 1.5 + 2 + 1.5 + 6 = 11 kW ; Latent = 1.1 kW

Total = 12.1 kW = 12.1/3.517 = 3.44 TR ; RSHF = 11/12.1 = 0.91

12. Solved Numerical Problems

Problem 1 : Bell–Coleman cycle

Air at 1 bar, 10 °C is compressed to 5 bar, cooled to 30 °C and expanded to 1 bar. Find COP and mass flow per TR ($c_p = 1.005$, $\gamma = 1.4$).

$r_p^{0.2857} = 5^{0.2857} = 1.584$; $T_2 = 283 \times 1.584 = 448$ K ; $T_4 = 303/1.584 = 191$ K

RE = 1.005(283 – 191) = 92.2 kJ/kg ; W = 1.005[(448 – 283) – (303 – 191)] = 53.8 kJ/kg

COP = 92.2/53.8 = 1.71 = 1/(1.584 – 1) ; $\dot{m}$ per TR = 3.517/92.2 = 0.038 kg/s

Problem 2 : VCR cycle (enthalpies from p-h chart)

$h_1 = 250$, $h_2 = 290$, $h_3 = h_4 = 95$ kJ/kg. Find RE, work, COP, flow rate and power per TR.

RE = 250 – 95 = 155 kJ/kg ; W = 290 – 250 = 40 kJ/kg ; COP = 155/40 = 3.875

$\dot{m}$ = 3.517/155 = 0.0227 kg/s ; Power = 0.0227 × 40 = 0.91 kW per TR

Problem 3 : Carnot COP for comparison

Evaporator −10 °C (263 K), condenser 35 °C (308 K): $COP_{Carnot} = 263/45 = 5.84$

Problem 4 : Fresh air and recirculated air mixing

See §7 example: $t_3 = 26.7$ °C, $\omega_3 = 0.016$ kg/kg.

13. Quick Revision & Important Questions

- $COP_R = Q_2/W$; 1 TR = 3.517 kW; Bell–Coleman: $COP = 1/(r_p^{(\gamma-1)/\gamma} - 1)$.
- VCR: $RE = h_1 - h_4$, $W = h_2 - h_1$, $h_3 = h_4$.
- $\omega = 0.622\, p_v/(P - p_v)$; $SHF = Q_s/(Q_s + Q_l)$; ET combines DBT, humidity, velocity.

→ Explain the Bell–Coleman cycle with T-s diagram. Derive its COP.

→ Explain the working of a vapour-compression refrigeration system with p-h and T-s diagrams.

→ List desirable properties of a refrigerant. What do R-12, R-22, R-134a, R-717 mean?

→ Define DBT, WBT, DPT, RH, humidity ratio. Describe the psychrometric processes.

→ Explain effective temperature and the comfort chart. How is cooling load

estimated?

~ End of Unit V ~